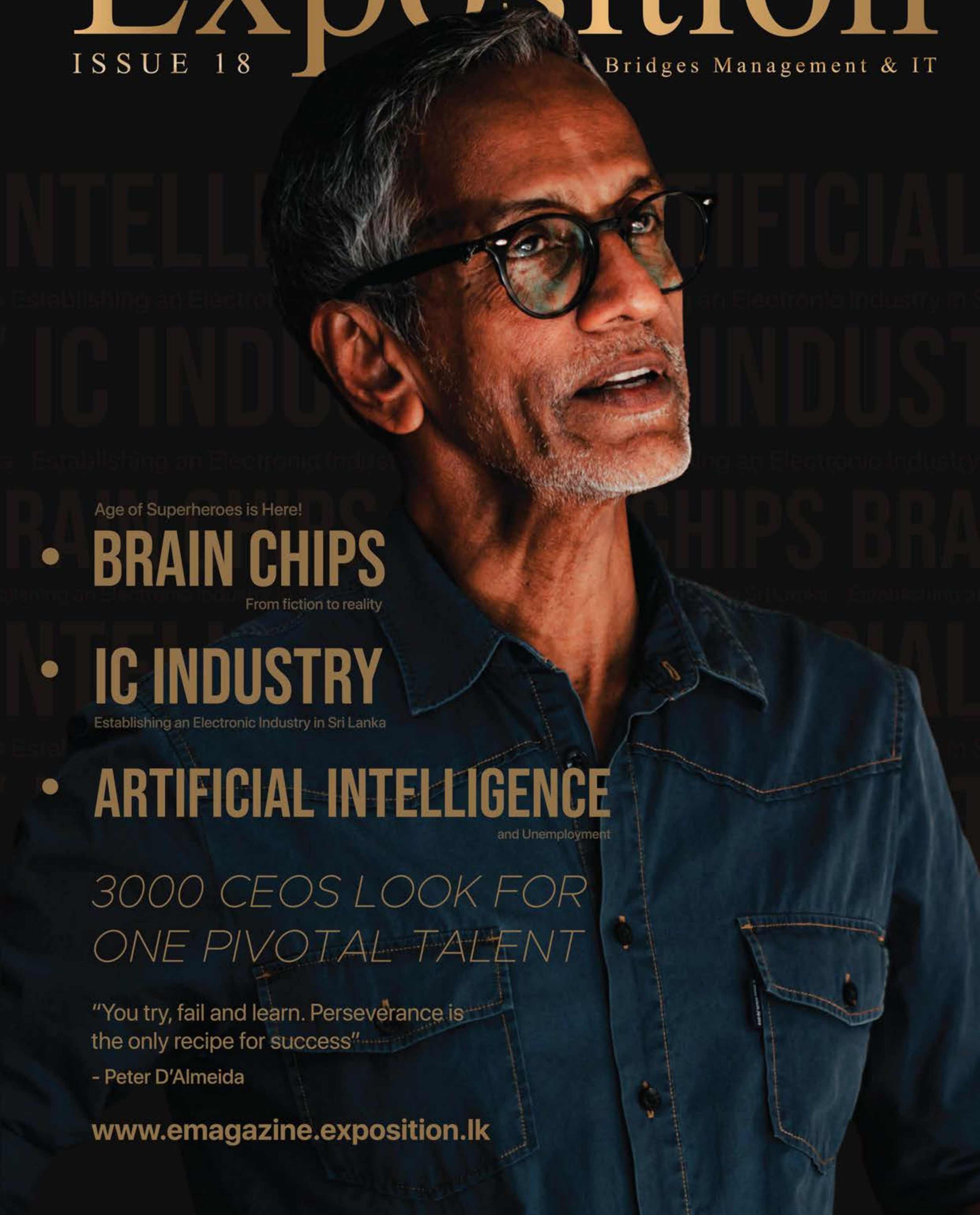
Exposition

ISSUE 18

Bridges Management & IT



Age of Superheroes is Here!

- **BRAIN CHIPS**
  From fiction to reality
- **IC INDUSTRY**
  Establishing an Electronic Industry in Sri Lanka
- **ARTIFICIAL INTELLIGENCE**
  and Unemployment

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- Peter D'Almeida

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Contact Us / Feedback:
Industrial Management Science Students' Association,
Department of Industrial Management,
Faculty of Science,
University of Kelaniya,
Sri Lanka.

Tel : +94-11-2903282, +94-11-2914482
Fax : +94-11-2903281
Email : imssa@kln.ac.lk
Website : mit.kin.ac.lk

Editors' Note

Encompassing the vision of "A Thought Leader in Digital Transformation," Exposition Magazine is launched by the Industrial Management Science Students' Association of the University of Kelaniya to bring forth the outstanding ideas of MIT undergraduates.

The Exposition Magazine is one of the leading university business magazines that aims to inspire the community through articles featuring novel IT innovations, the latest business trends, interviews with eminent personalities, and new research-based information.

We are honored to be the editors of Exposition Issue 18, and are grateful for the support and guidance received from all those who contributed to this publication. Our heartfelt gratitude goes out to the academic and non-academic staff of the Department of Industrial Management, the IM alumni, and our colleagues for their invaluable support. We would also like to acknowledge the hard work and dedication of the previous Exposition teams for paving the way for us to carry on the legacy of this publication.

In addition to the magazine, we are proud to have successfully organized "hackX," an inter-university startup challenge, and "hackX Jr.," an inter-school hackathon, for the seventh and fifth consecutive years, respectively. We were also able to continue "Edify," the inter-university article competition, for the second successive year. These events continue to grow in popularity, and we are confident they will continue to raise the prestige of the Department of Industrial Management to greater heights.

As undergraduates, we are entrusted with sharpening our skill sets to advance in our future careers. Exposition is one such venture for us to engage with the corporate world and develop our potential as thought leaders in digital transformation. This magazine includes a remarkable collection of articles and features that reflect the vibrant and dynamic community of the Department of Industrial Management.

We invite you to join us in experiencing the legacy of Exposition Issue 18, and we hope that you find the contents both informative and inspiring.

Disclaimer

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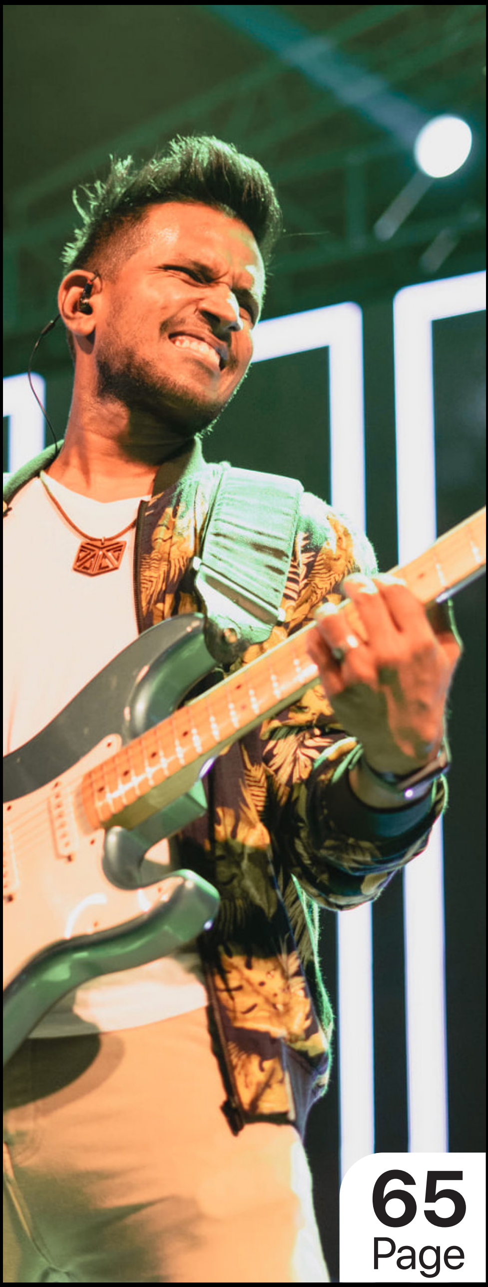
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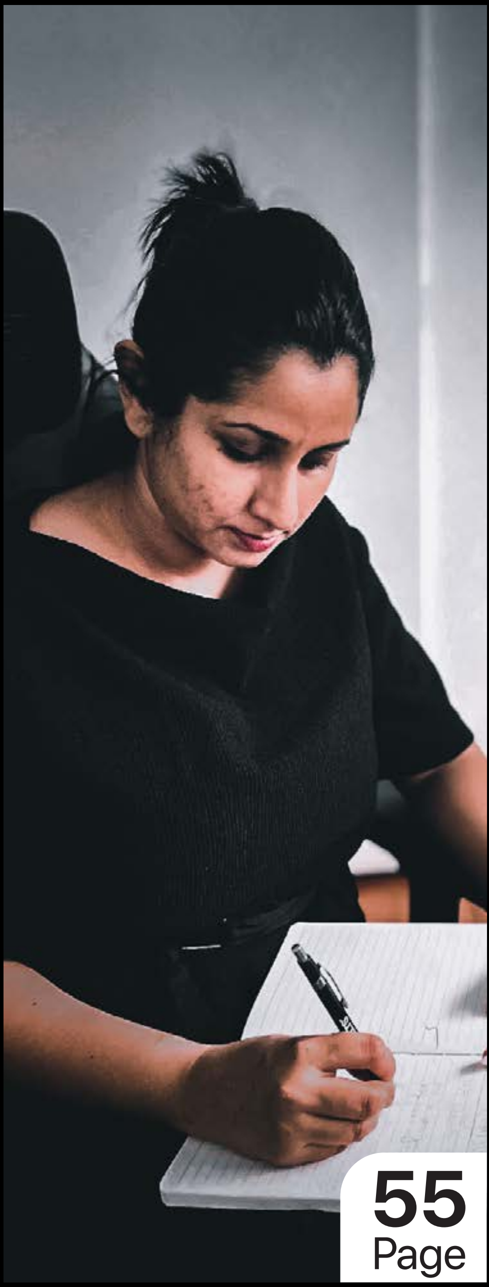
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UOK

University of
Kelaniya.

The University of Kelaniya is one of the premier state universities established in Sri Lanka. Located in Kelaniya, just outside of Colombo, it is home to two institutes, twenty seven centers, seven faculties and fifty-five departments.

The University of Kelaniya originates from the historic Vidyalankara Pirivena, founded in 1875 by the Venerable Ratmalane Sri Dharmaloka Thero as a center of learning for Buddhist monks. With the establishment of modern universities in Sri Lanka, in 1978 the Vidyalankara Pirivena ultimately became the University of Kelaniya. The university has the vision "To become a center of excellence in creation and dissemination of knowledge for sustainable development" and works towards its mission which is "To nurture intellectual citizens who can contribute to national development, through creativity and innovation". It has several departments not found generally in the Sri Lankan University system.

The university provides students with the finest facilities in all aspects; the Academic Division, Examination Division, Medical Center, Library, Student Welfare Division, Career Guidance Unit, Physical Education and Research Council are significant among them. It has unique fields of study that has great potential to attract both local and international students. It has the highest number of international students, the number exceeding 500 and offers over 100 scholarship opportunities annually.

Department of Industrial Management.

Since its inception in 1967, the Department of Industrial Management, Faculty of Science, University of Kelaniya, has proven itself a pioneer in teaching management and IT to science students in the Sri Lankan university context. It was established with the aim of molding science students into professionals who are well versed in both management and IT streams. Currently, the department offers two degrees, a Bachelor of Science (Hons) in Management and Information Technology (MIT) and a Bachelor of Science (Hons) in Information Technology (IT). Apart from theoretical knowledge, both degree programs also embrace the development of soft skills required for graduates to be successful in the professional world.

For those interested in qualifying further, the department also offers a Master of Information Technology and a Master of Data Analytics. The Master of Information Technology program is designed for graduates desiring to enhance their competencies when pursuing a career in Information Technology, and for those graduates in other functional areas desiring to use Information Technology to enhance their ability to provide effective and efficient solutions to their stakeholders. The Master of Data Analytics program is designed in such a way that the candidates are given the opportunity to start from the basic level and then advance their competencies in using cutting-edge tools and technologies in diverse domains of applied analytics.

Equipped with modern facilities, state-of-the-art technology, and an international theological library, the Department of Industrial Management is the best place for those seeking proficiency in management and Information Technology.

The student body of the department, the Industrial Management Science Students' Association (IMSSA), provides undergraduates numerous opportunities to develop their leadership, communication, problem-solving, and decision-making skills through the successful execution of projects such as guest lectures, InfoStudio sessions, Smiling Mindz sessions, and field visits. Additionally, it provides members with industrial exposure by arranging events such as hackX, hackX Jr. and Exposition.

In recognition of its journey to excellence, the Department of Industrial Management was awarded the ISO 9001:2015 Quality Management System Certificate in 2020 for the 3rd consecutive time. The ISO 9001:2015 quality certification ensures institutes and entities are performing to the optimal standard for continued prosperity and improvement.

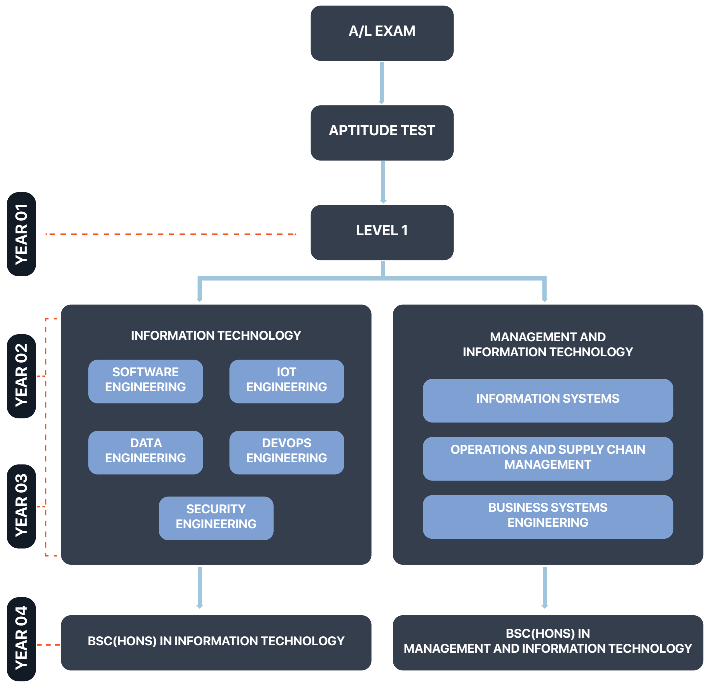
MIT.

Management and Information Technology

The Department of Industrial Management, one of the first-ever departments of the Faculty of Science, University of Kelaniya, introduced the B.Sc. (Hons) in Management and Information Technology intending to bridge the gap between academia and the corporate world and meet the demands of the industry.

B.Sc. (Hons) in Management and Information Technology is a four-year honors degree program that offers a wide range of knowledge and skills as a blend of Management and IT. Apart from the subject knowledge, this program emphasizes soft skills and personality development, including an internship opportunity for six months and the option to exit with a three-year general degree upon the approval of the faculty.

The students following this degree program will get the opportunity to further specialize in one of the following areas from their third academic year onwards; Business Systems Engineering, Information Systems, and Operations and Supply Chain Management.



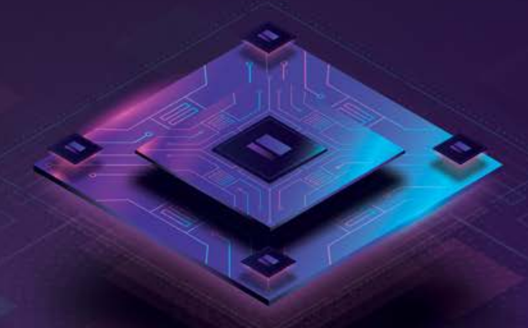
The department also offers a B.Sc. (Hons) in Information Technology starting from the academic year 2019/2020. The subsequent three years of the B.Sc. (Hons) in Information Technology degree program will focus on building the Information Technology competencies of the students, depending on their career objectives. The program considers the increasing national and international need for computing professionals, and has designed a curriculum that follows the latest guidelines of recognized professional bodies such as the ACM and IEEE. The undergraduates can choose a career in Information Technology specializing in Software Engineering, IoT Engineering, DevOps Engineering, Security Engineering, Data Engineering or Data Science.

The blend of the Management and Information Technology fields is an advantage for graduates in rising the career ladder. They enjoy a high level of employability due to their reputation for high potential in the working capacity.

Students wishing to join the B.Sc. (Hons) in Management and Information Technology and B.Sc. (Hons) in Information Technology degree programs should apply through the University Grants Commission with relevant results and a Z-score in the Physical Science or Biological Science stream for the G.C.E Advanced Level examination. In addition, they should pass the aptitude test conducted by the Department of Industrial Management.

The program aspires to inspire future undergraduates to set their own goals and discover a brighter future.

ESTABLISHING AN ELECTRONIC INDUSTRY IN SRI LANKA



By **Dr. Shan Jayasinghe** | Senior Lecturer

Integrated Circuit (IC) has become a hot topic in international media because, at present, the world is experiencing a severe shortage of ICs. This has adversely affected the production capacity of many high-tech manufacturers since the functioning of each and every high-tech product depends on the processing power of various types of ICs.

Figure 1 : Made in china 2025 - Target Sectors

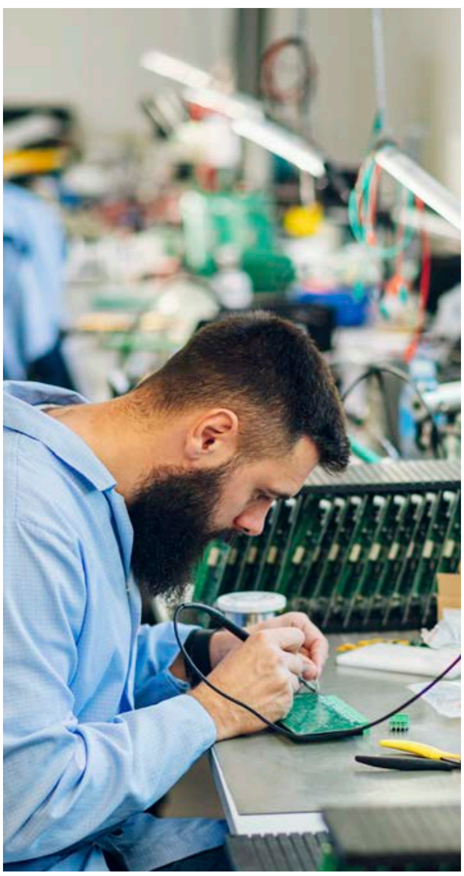
At the same time, using IC as a weapon, the United States of America (the USA) is in a technology cold war with China. At present, the US is the leader in the global IC industry. There are two main components in the global IC value chain, namely, IC design and IC manufacturing. The US is the leader in IC design. Taiwan, which is an ally of the US, is the leader in IC manufacturing. Furthermore, most of the sub-components of the global IC value chain are positioned on the shores of allies of the US (e.g. roughly 70 percent of etching gas and 90 percent of photoresists for the world market are produced by Japanese companies).

> **“Chip shortage spirals beyond cars to phone and consoles.”**
>
> *by Debbie Wu, Vlad Savov and Takashi Mochizuki*
> *6 February 2021, 07:00 GMT +11*

In 2015, China started its Made in China 2025 (MIC2025) initiative to become the world leader in 10 industries (please see Figure 1). Out of the 10 industries, nine depend on the processing power of ICs. Apart from that, in China, there are many other industries that depend on the processing power of ICs. China imported ICs worth USD 320 billion in 2018.

The US considers MIC 2025 a threat to its technological dominance. Therefore, to block MIC 2025, the US decided to use IC as a weapon. In March 2020, the US ordered the world’s largest IC manufacturer, TSMC, to terminate its supplies to Huawei Technologies, China. Furthermore, the US requested that its IC firms move their manufacturing plants out of China (e.g. GlobalFoundries scrapped its USD 4 billion investment and moved out of China). China received this as a killer blow to its MIC2025 initiative and the survival of many other tech industries. Beijing realized the significance of developing its own semiconductor industry. Therefore, the government of China initiated a project to become self-sufficient in ICs. The aims of the new initiative are to fulfill 70% of the IC requirements of China by 2025, to become a major exporter of ICs by 2025, and to become a world leader in every aspect of IC technology by 2030. The Chinese Communist government has pledged that they will pour a massive USD 1.4 trillion into high-tech industries until 2025, backing companies such as chipmaker SMIC.

Following the path of the US, China might also aim to position its own IC value chain within its allies to establish its geographical presence. By observing the trends in the US dominant value chain, it is possible to assume that China might be willing to position various medium and low-end manufacturing operations of the IC value chain, such as wafer manufacturing, assembling, and testing, outside of China.

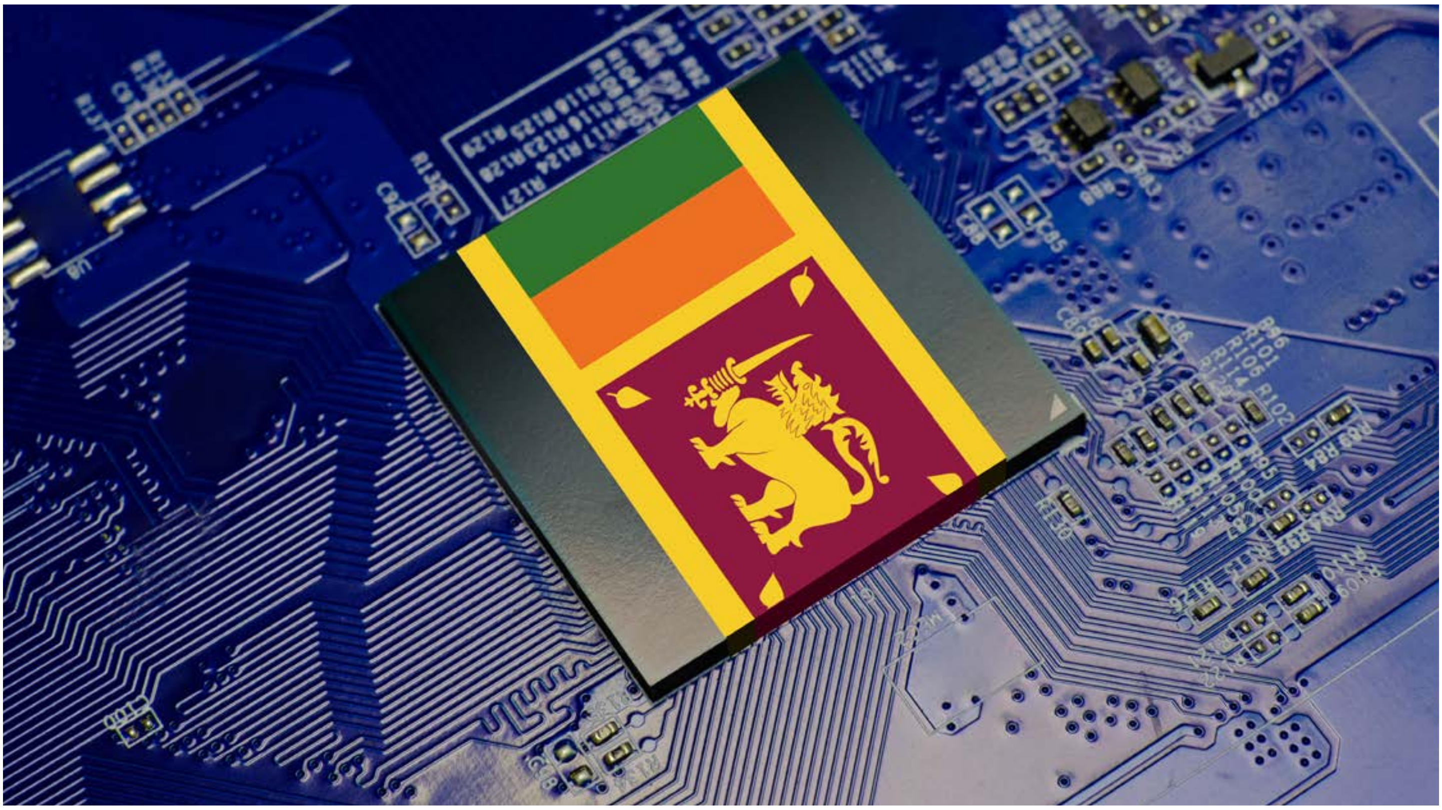


Based on this background, the IC shortage and the geo-political trends created by the trade war between the US and China have created a window of opportunity for countries such as Sri Lanka. When shifting from low-end agriculture-based economies to high-tech manufacturing economies, countries such as Taiwan, South Korea, Malaysia, Singapore, and the Philippines have experienced direct government intervention to attract initial Foreign Direct Investments (FDIs). For example, Malaysia received its first high-tech manufacturing facility from Intel Inc. back in 1972 as a result of the bilateral trade discussions between the USA and Malaysia. Since then, Intel Inc. has invested about USD 6.5 billion in the country over the past 48 years. More recently, the Vietnamese government invited Samsung Semiconductors to develop a USD 1 billion IC manufacturing plant in Ho Chi Minh City through the South Korean government.

Going by this trend, if Sri Lanka is to establish a high-tech industrial culture within the country by capitalizing on the above-mentioned window of opportunity, the direct policy and diplomatic involvement of the government are a must. As discussed earlier, the IC value chain consists of two main components, namely, IC design and IC manufacturing. Initially, Sri Lanka should try to establish an IC manufacturing operation in the country. Sri Lanka could approach the US, Chinese, and/or South Korean governments for assistance. Based on the socio-cultural situation in Sri Lanka, it is ideal if the country can attract investment for an IC assembly and packaging plant. Ideally, this should happen by the end of 2023. Once the assembling and packaging operation is up and running, the

government should focus on attracting IC wafer manufacturers to set up IC foundries. However, the establishment of a foundry is not as easy as establishing an assembly and packaging plant due to the technological sophistication and the requirement of higher volumes of water and uninterrupted power. At the same time, the government should directly intervene to promote Sri Lanka as a high-tech investment destination for consumer electronics companies. Sri Lanka is an ideal investment destination for Chinese, Taiwanese, South Korean, and Japanese consumer electronics companies. Ideally, the government should focus on establishing Sri Lanka's first foundry and/or consumer electronics manufacturing facility by the end of 2028.

When considering the long-term sustainability of a high-tech industrial culture, focusing only on the manufacturing aspect of the industry is not sufficient. Countries such as Malaysia and Thailand are focusing mainly on the manufacturing aspect.



Malaysia is already caught in the middle-income trap, and it is forecasted that Thailand will also be caught in the middle-income trap due to an overreliance on high-tech manufacturing. When studying countries such as Taiwan and South Korea that have successfully evaded the middle-income trap, it is evident that any country that plans to establish a sustainable high-tech industrial culture should focus on developing a human resource that can contribute toward the advancement of the global high-tech intellectual value chain. Hence, we propose to the Sri Lankan government to start a degree program specializing in IC design starting in 2023. The government has the option of establishing this degree in one of the existing engineering faculties.

If the government is successful in establishing an IC assembling and packaging facility in Sri Lanka by the end of 2023, the graduates who will enroll in the degree program could get themselves trained in the IC assembly and packaging plant. Once they graduate, they should be provided with special training in IC design at a research center that is dedicated to high-tech research. Ideally, this research center should be backed by a global player in IC design. The best option Sri Lanka could think of is HiSilicon, the IC designing arm of Huawei Inc.

Sri Lanka has been relying on low-end exports such as foreign remittance, tourism, agriculture products, apparels, etc. to generate export revenue. However, none of these exports is capable of providing the massive boost the country needs if it is to become a USD 250 billion economy by 2035 (as mentioned in the president's manifesto). At present, Sri Lanka has a USD 88 billion economy. To become a USD 250 billion economy, the country has to add USD 162 billion to its economy. Ideally, exports should account for a considerable portion of that amount. However, if the country continues to depend on the above-mentioned exports, it will be impossible to receive a considerable contribution from them. This is another reason why Sri Lanka should try to capture a portion of the global high-tech value chain.

Dr. Shan Jayasinghe
Senior Lecturer,
Department of
Industrial Management,
University of Kelaniya


What you need to do is start from somewhere & grow through it.
INTERVIEW
Dhanushka
FERNANDO

"Life is a process, not a product."

Entrepreneurs constantly give power to this process by propelling it through their passions. With entrepreneurs like Mr. Dhanushka Fernando, this life process influences others to be very passionate and make any industry where it needs to be, securing maximum profitability and longevity. Mr. Fernando believes that entrepreneurship can play a vital role in income generation and the revival of the economy. It is with great pleasure that Exposition Issue 18 presents Mr. Dhanushka Fernando, hoping his exemplary personality will inspire undergraduates and others to pursue their dreams.

Q: How did your upbringing and formative years influence who you are today?

It all goes back to my school and college days, where the exposure and motivation, both from my parents and my school, taught me to be independent and perseverant at a young age. I played sports and even captained some of our school teams. Also, the belief that 'knowledge is very important' was inculcated in me at the time. It was those early days that made me who I am today, and I would like to thank my parents and my school St. Thomas' College, Mt. Lavinia for molding me into a "total personality."

Q: How has the volunteer work you were involved in at Red Cross led you down your current path?

While I was reading for my degree, the Tsunami struck Sri Lanka. I noticed the need to support the people affected by the tragedy. The Red Cross was doing a lot of projects related to construction, healthcare, education, water, and sanitary care. I got the opportunity to volunteer with them. That made me understand the vulnerabilities of our fellow citizens. Red Cross gave me a way to give back to the country and encouraged me to stay in Sri Lanka. Initially, I did not have the financial capability to support those in need but through Red Cross, I was able to join in on projects and achieve self-satisfaction that I am doing something for our community.

Q: What compelled you to start a firm specializing in marketing and then later focus on furniture, when you began your undergraduate journey in Information Technology?

My first degree was in Information Technology. But after a couple of years, I realized that I am better at PR and sales. So, I did my CRM certification and continued my studies in marketing.

While studying, I wanted to get some work experience, so I started off with Red Cross. There, my role was associated with communication and project management. Afterward, I joined an online marketing company. Today, marketing and technology go hand in hand. But around 10 to 15 years back, these were two different subjects. In 2010, I noticed an opportunity in the industry for a newcomer in the digital marketing and technology solutions provider domain. The background I got from learning IT helped me immensely to master the technological part of marketing and make a name in the industry.

Q: Could you explain what "FINEZ" is, and why you started it?

My father was in the furniture industry. And having seen the works at a young age, I thought of venturing into the field. So, we opened a new factory called "FINEZ" in 2015. There was also a boom in construction at the time, and I noticed the demand for furniture. Instead of going the conventional route, our business model prioritized quality, design, and delivery. We wanted to differentiate ourselves from the other players in the market. Now we are more of a home decor retail brand, offering other products and solutions in addition to furniture.

Q: Since the beginning, how have your priorities evolved?

Those in their 20s often get pressured to do a degree, find a job, get married, build a house, and so on. Youth should get a chance to pick their path. What you have

experienced during your 20s enables you to apply what you have learned and make better decisions when you are in your 30s or 40s.

I took a bold step in my 20s to venture into entrepreneurship when people were not too open to the concept. With experience, ups, and downs, my priorities have evolved from establishing a business to sharing knowledge and experiences, guiding and motivating the younger generation to learn from their mistakes, be courageous and do what they like.

Q. What were the obstructions along the way, and how did you overcome them?

There are two obstacles I have identified. First is the obstacle created by my own mental barriers. Feeling demotivated and stressed is tough. There are days you might feel discouraged. It is necessary to talk with someone, especially with people who have already done what you are doing, and learn from them.

Next are the environmental factors-political, economical, social factors that could affect a person. Even in the last couple of years, there was COVID, and a lot of businesses went through the pressure of how to pay their team and keep the business functioning. That is where it is important to diversify or be innovative. My experience is, if a business can innovate ahead of the market changes, they are always able to stay ahead of the curve.

Q. You are more than just an entrepreneur. You are a social activist, an athlete, and a consultant. How do you structure your life to manage all of these activities?

People need to recognize the importance of nurturing both their mind and body. If you want to see a change in your body, mind, or career, you need to act now so you can see the results after a year. These should be taught. Self-care is important. There is a saying that some people die in their 20s and get buried in their 70s. You should not let this happen. Just make an effort, even the little things like dressing smart can go a long way. Then comes the learning part. You should seek knowledge, to not only get a job but to become a better person. The more you read, and the more you learn, the more balanced you become. Another thing is to discourage negativity because it is through positivity and optimism that you always try to find a way. Once you nurture yourself, then only you can give back to the community. One should always try to give their best version at all times, irrespective of what happens around them.

Q: You have amassed many accolades and respect in the industry as a successful entrepreneur. What achievements make you the proudest, and what do you believe is the driving force behind your success?

When I started as an entrepreneur, I had the freedom to do what I wanted to do. Being my own boss, the sort of satisfaction and freedom I felt was immense. Another achievement was getting the right type of people and seeing them succeed in what they do. When seeing people achieve things and grow into their roles, I feel proud. The other satisfaction is like planting a tree and seeing it come to fruition. It is not only about making money. Providing employment, giving support, and seeing customer satisfaction, add value to society and also to the community. Yes, awards bring recognition and validity, but more than that, being a value-adding entrepreneur and encouraging others to do the same are my biggest achievements in terms of myself. If you condition your mind to be satisfied with those achievements, you won't go wrong as an entrepreneur.

Q: Everyone understands how difficult it is to start a business. If one wishes to invent a new idea, many risks will be involved. As a result, young people frequently have feelings of self-doubt and fear of failure. How did you handle your anxiety and skepticism when you first started?

For about 5 to 6 years, I worked for corporates. I wanted freedom, I wanted to get away from that culture, and it was more towards doing my own thing, and being my own person. I didn't have a lot of anxiety in terms of taking that risk since I wanted to do so. I had some savings with me and it was enough for me to work on. Initially, it was only me, but then I hired a couple of designers and someone to coordinate. Financially we were successful. But then we moved our office to Colombo and there were a lot of overheads like traveling costs. So from around 2014-2015 the "real business process" started. We were expanding, hiring, so there was a lot of stress and pressure. It was a learning curve, from starting a business to registering the company, to opening up the bank accounts. I had to learn the hard way, with no one to guide me through the process. So if someone wants to start as an entrepreneur, I would suggest getting a mentor. At the same time, it is important to develop a mindset, to not give up, and to learn from failures without being afraid to make mistakes.

Q: As a consultant, what insights would you share with entrepreneurs looking to break into the field?

Entrepreneurship is about finding opportunities and solutions to solve problems. In the current market dynamics, the available opportunities and solutions that can be given, change very fast. It is important to understand that because what trends today, might be forgotten next year. Also, after finishing school or graduating, you have sound knowledge in certain areas. It is easier to venture through those areas as you have a better understanding of that field. If you want to do something completely different from what you have studied, then it's important to put forward your skills and knowledge. Entrepreneurship revolves around dealing with people and finding solutions, selling them to people, understanding their psychology, and why they would buy your product or service. You have to be creative and innovative. And in this day and age, you can use social media platforms to reach a wider audience.

Q: In your opinion, what are some business prospects that fresh graduates would find worth exploring?

The main thing is to focus on an area where they can generate revenue outside of Sri Lanka. Our market conditions are tough, and it is difficult to enter the industry, so that is the best option. There is a lot of opportunity for local products to earn foreign income. Then there are freelancing platforms where people can offer their skills at a dollar rate. As long as you have a computing device and an internet connection, you can offer these services. So focus on that. These avenues will guide you to become an entrepreneur. What you need to do is start from somewhere and grow through it.

EEC | Exposition Entrepreneurial Community

The Exposition Entrepreneurial Community, introduced by Exposition, is a platform to bring together entrepreneurs in different fields of the industry in Sri Lanka to help discover and discuss new ideas.

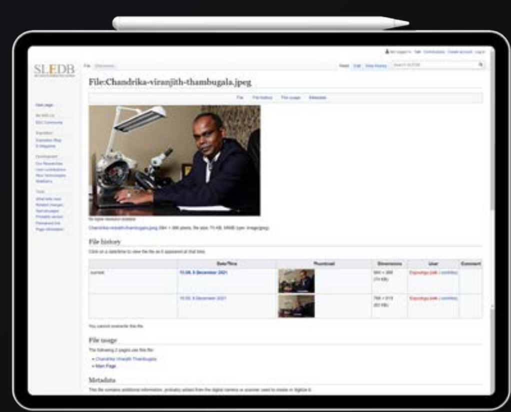
The main stakeholders in the EEC are the small and medium-scale entrepreneurs in the island. These entrepreneurs are innovators and risk-takers. They are the sources of new ideas, which evolve into goods, services, or business procedures. They play a vital role in the economy by contributing their skills to develop and uplift the industry and economy.

The EEC consists of two main phases:

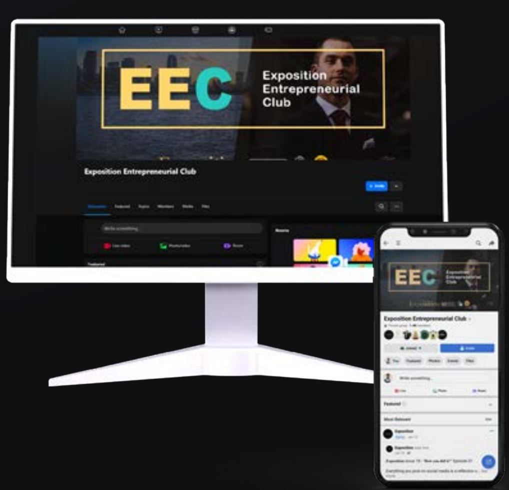
- The Online Community Forum
- Sri Lankan Entrepreneurial Database

SLEDB - Sri Lankan Entrepreneurial Database

SLEDB is a knowledge base that consists of information and profiles of SME sector entrepreneurs. The main objective of the SLEDB is to create a linkage between material suppliers and entrepreneurs in the SME sector. Entrepreneurs can gather considerable knowledge on maintaining a business, managing resources, and the challenges of sustaining a business through EEC. Given the number of entrepreneurs who do not have a notable IT or Management background, we MITians have recognized it as our responsibility to facilitate their requirements and assist them in becoming successful. We hope to form alliances with various organizations in order to meet the needs of entrepreneurs in the SME sector. Additionally, entrepreneurs themselves get the chance to know each other and create links between themselves. The EEC will provide excellent opportunities for SME sector entrepreneurs to grow their businesses in the future.



The Online Community Forum



The online community links the entrepreneurs of the community while opening doors to exchange new ideas and knowledge among experts and enthusiasts in the industry. The online community is a knowledge pool that facilitates all the entrepreneurship enthusiasts and passionate entrepreneurs out there to expand their horizons by exchanging expertise within the community. Currently, this platform is operating as a Facebook group, which consists of over 1400 community members. This is the best platform to explore new trends and achieve the next best level of their business. It provides consultation via virtual brainstorming sessions conducted by the industrial experts who have already joined hands with our online platform.

INTER - UNIVERSITY STARTUP CHALLENGE

Following the success of three consecutive hackathons and three highly successful startup challenges, hackX 2022 is the 7th resplendent iteration of the inter-university startup challenge organized by the Industrial Management Science Students' Association, Faculty of Science, University of Kelaniya. With the theme of "Sustainable Innovation," hackX 2022 was not only limited to tech-based solutions but also open to any sustainable, eco-friendly idea that will help planet Earth survive. hackX aims to establish a platform to expand passionate developers' innovative ideas into real-world sustainable solutions and earn due industrial recognition and support for their innovations.

hackX 2022 consisted of 3 phases: ideaX (Initial idea pitching event), designX (Startup guidance program) and the hackX 2022 final event. hackX 2022 attracted over 200 proposal submissions, of which 30 were shortlisted to compete in the semi-finals. ideaX, the semi-finals of hackX 2022 was held on the 25th of September 2022 at the Department of Industrial Management. The teams got the opportunity to pitch their innovative ideas before a distinguished panel of judges, composed of Mrs. Harshani Ranawaka, Ms. Shyama Gunarathne, Mr. Deenadayalan Nagaratnam, Mr. Pasindu Bandaranayake, and Mr. Hasitha de Silva. Based on the ingenuity of the idea, its feasibility, and its potential, 12 teams were selected to compete at the Grand Finals. Ideasprint, the intra-department startup challenge, was held in parallel to select 3 teams from the Department of Industrial Management to compete at the grand finals.

designX comprised 4 webinar sessions conducted to develop the skills of the finalists. Topics such as Optimizing cloud costs, Turning ideas into reality, Enhancing your product with an API and a Roadmap to Entrepreneurship were addressed during the sessions.

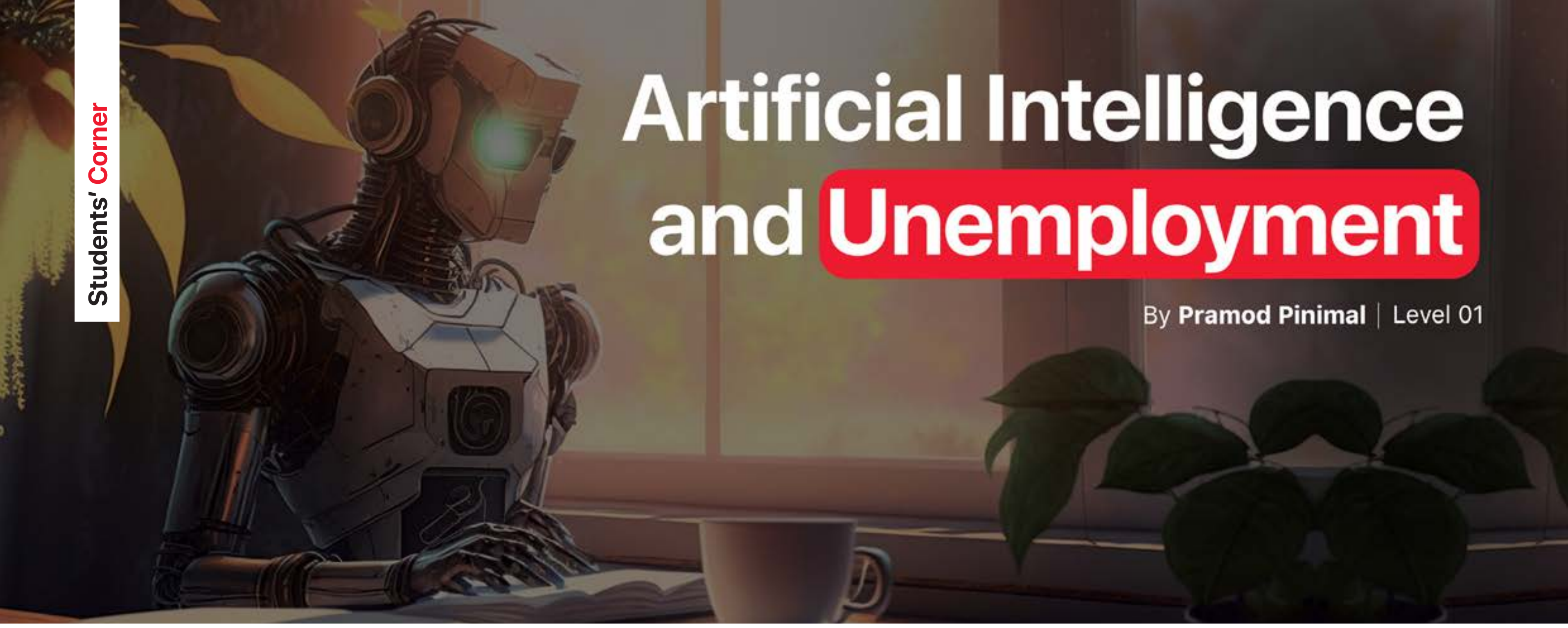
The Grand Finals of hackX 2022 were held with much grandeur on the 20th of November 2022 at Ideamart, the auditorium of Dialog Axiata. Preceded by a few encouraging words from Professor Janaka Wijenayaka, Head of the Department of Industrial Management, the finalists took the stage. They demonstrated their developed products to the judges.Team IdeaLabs of the University of Sri Jayewardenepura became the champions of hackX 2022 with their product "NeuroX," a device for identifying neurological disorders using line drawing techniques. Team Xentrix from the Sri Lanka Institute of Information Technology and Team Zyndicate from the University of Kelaniya emerged as the first and second runners-ups, respectively. The three winning teams received valuable cash prizes and gift packs from Sana Commerce Sri Lanka. Dr Chathura Rajapakse, Mr Ruwanthaka Ranasinghe, Mr Tuan Cassim, Ms Amaya Jayasuriya, and Mr Deenadayalan Nagaratnam comprised the honorable panel of judges at the Grand Finals. Corporate sponsor representatives, lecturers, and undergraduates also added color to the event.

As a special segment of hackX 2022, the "Most Popular Team' was introduced. Team Perpova from the University of Moratuwa emerged as the Most Popular Team and received valuable gift packs from Connex IT.

A notable mention should go to the official partners of hackX 2022, whose contributions led to the success of the event: Official Main Sponsor Sana Commerce Sri Lanka, Official Co-Sponsors Creative Software and Connex Information Technologies, Official Talent Partner Evonsys (PVT) LTD, Official Venue Partner Ideamart, Official Innovation Partner Loons Lab, Official National Partner Information and Communication Technology Agency of Sri Lanka (ICTA), and the Official Mentoring Partner 99x.

Waruna Sri Wickramasinghe and Thakshanah Selvakumar were the chief coordinators of the event. Team hackX also included Manod De Silva, Vishaka Bandara, Udara Suranimala, Tharindu Weerathunga, and Kavisha Amarasinghe from Level 2. The event would not have been a success without the immense support extended by the lecturers, academic and non-academic staff members, and undergraduates of the Department of Industrial Management.

www.hackx.lk



Artificial Intelligence and Unemployment

By **Pramod Pinimal** | Level 01

The idea of technological unemployment is making a comeback, both in the labor economics literature and in the public, underscoring concerns that job destruction could result from technology advancement. This fear stems from the belief that production will be relentlessly automated as a result of exponential technological advancement.

Robotics and artificial intelligence (AI) are two relatively new computer science subfields that have fueled this ambition. The first builds on earlier technological revolutions that have occurred throughout history and attempts to lessen laboriousness, advance technological frontiers, and boost industrial effectiveness. The invention of printing is credited to Gutenberg in the mid-15th century, despite evidence that Tang Dynasty China had already mastered printing techniques by the 8th century, and modern printing presses have contributed to accelerating the diffusion of information by speeding up printing while decreasing costs, from manual loaves of parchment to the modern cranes that build ever taller skyscrapers. Robotics contributes to this continuity by drastically reducing the amount of human labor required to operate each unit of capital by automating machine operation.

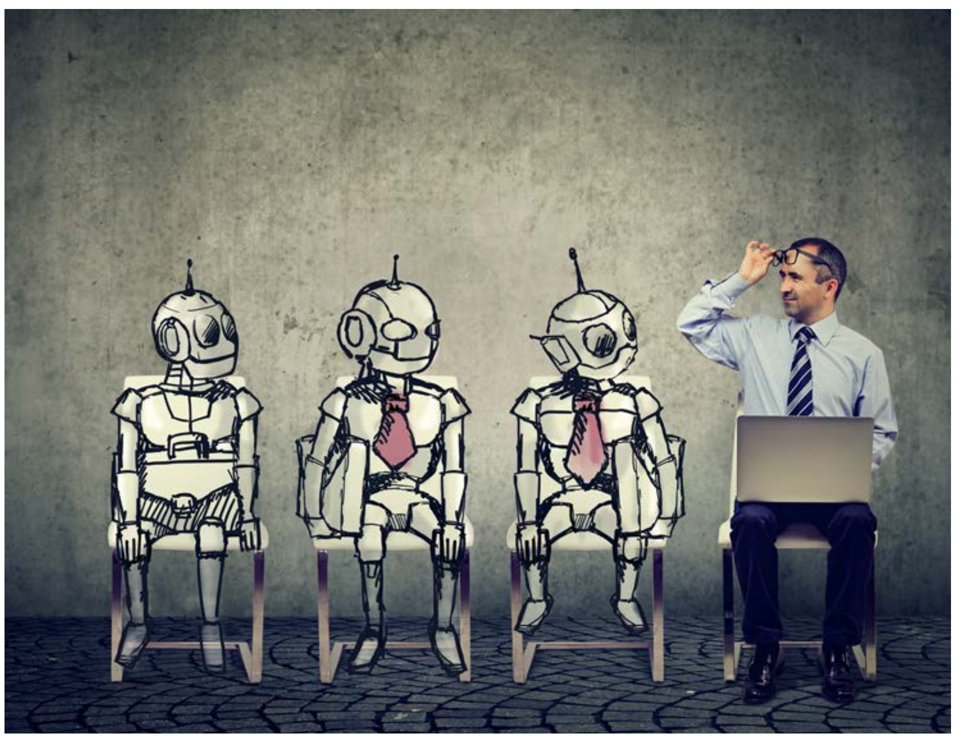
The pursuit of efficiency is shared by artificial intelligence, but it varies from earlier advancements in that it does not automate cognitive processes rather than manual ones. In this way, the development of computer science, which enables speedy and simple computing of tasks, is a continuation of the digital revolution. In contrast to traditional algorithms, which are static, artificial intelligence (AI) differs in that it is adaptive. A machine learning algorithm is programmed to learn how to accomplish a task, in contrast to a standard algorithm, which is programmed to perform a task. When a task isn't very complex, the first method is far more effective and uses up less time and resources. The second way is significantly more potent, though, since the level of complexity rises and occasionally produces a consequence that the creators did not intend. This kind of algorithm works especially well for issues requiring tacit knowledge, which is by nature challenging to document. For instance, the development of driverless vehicles has advanced significantly thanks to machine learning techniques. Driving is a straightforward activity, but it requires a high degree of adaptability, which was exceedingly challenging to achieve with conventional algorithms because it

would have required foreseeing every scenario the car would encounter.

This research seeks to respond to a single, straightforward question: Is there actual evidence that the use of robotics and AI technologies can lead to higher unemployment? Because the total result of employment creation and destruction is what counts to us in this situation—that is, determining whether compensatory mechanisms are effective—we want to reason at the macro level. The classical and Keynesian Schumpetarian compensation mechanisms were set out by Calvino and Virgillito. The former includes four different mechanisms: an increase in labor demand in the companies producing the process innovations, a decline in prices as a result of productivity gains that then causes an increase in aggregate demand, a decline in wages that also increases labor demand, and an increase in investment funded by the drop in the unitary cost of production brought on by process innovation that prompts new hiring and increases production. The second category contains two mechanisms: the introduction of new products that open up new markets and hence raise labor demand, and an increase in pay due to productivity gains for the workers who are still employed by the company.

Optimistic and pessimistic perspectives are both present in discussions on technological unemployment. In order to prevent technical unemployment, Keynes advocated for a reduction in working hours, stating that "three-hour shifts or a fifteen-hour week may put off the problem for a great while." Even further, Rifkin predicted the end of employment, imagining an almost totally automated economy that would generate an abundance of products and services at a marginal cost that was almost zero.

They discovered that certain professions saw slower employment development than other occupations, and occasionally even a slight fall in employment levels. The authors draw attention to the fact that overall, automation does not lead to a net loss of jobs. Artificial intelligence has little influence on employment, but it does have a slight positive impact on earnings, according to Felten, who proposed the "AI Occupational Impact" measure. Acemoglu analyzed information on online job postings and discovered no link between exposure to AI and employment or pay. Webb, who found that jobs with high exposure to automation technology witnessed a fall in employment and income, disagrees with this finding. Felten, and Frey and Osborne are built upon, Bessen's investigation into the relationship between jobs and automation at the organizational level came to the conclusion that automation increases the risk of employees leaving an organization. However, they point out that this impact is minor and gradual. In their examination of a sample of Spanish manufacturing companies, Koch found that the adoption of robots resulted in a 10% net growth in

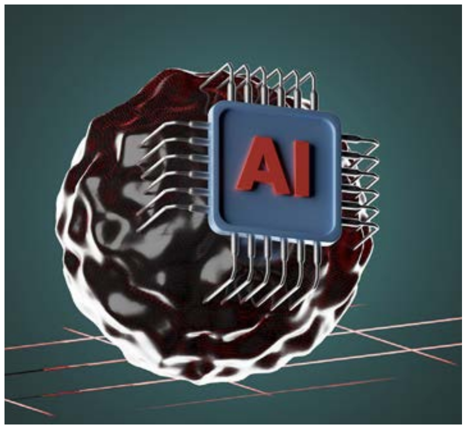


jobs, whereas companies that did not invest in robots had job losses over the same time period. Using French data, Aghion discovered that automation has a favorable effect on employment at both the business and industry levels. Montobbio emphasized a sector-specific impact on manual and cognitive activities by comparing labor-saving robots patents with firm-level statistics. There is a vast body of work that discusses the effects of process improvements on employment in addition to studies on robots and artificial intelligence.

In contrast to earlier studies that showed no indication of a significant relationship between the two, Van Roy's research found that process innovation tended to enhance employment.

While firm-level analysis has the advantage of giving a lot of data, frequently of higher quality than macroeconomic data, it also has the drawback of not offering a useful framework for determining the overall net impact of technological advancement on employment. Indeed, the creation of new jobs in other firms or sectors may be able to counterbalance certain job losses in one company or industry. They did, however, make note of the correlation between rising robotization and a decline in the employment percentage of low-skilled individuals. Acemoglu and Restrepo examined the impact of robot density on local labor markets in the US using a task-based theoretical framework and discovered that each additional robot per thousand workers is associated with a 0.2% decline in the employment-to-population ratio and a 0.37% decline in wages. By demonstrating that there is a bigger negative impact on jobs and hourly wages the less educated the workers, they further highlighted the heterogeneity of the consequences.

We develop four research hypotheses based on empirical and theoretical literature. First, we



anticipate finding a link between the overall unemployment rate, robots, and AI (H1). While some writers have emphasized a positive influence on employment, the quantitative impact of AI on jobs at the macro level is currently understudied. Second, we anticipate a positive relationship between robots and the unemployment rate of workers with the lowest levels of education and a negative relationship between robots and those with higher levels of education. According to the skill-biased technical change (SBTC), skilled employees profit from technological advancement while unskilled ones suffer negative effects. On the other hand, we examine the claim that AI and robotics primarily affect the unemployment rate of people with a medium level of expertise (H3). This is in reference to the literature on labor market polarization, which claims that AI algorithms primarily replace routine tasks carried out by medium-skilled workers.

Finally, we anticipate varied impacts within age groups and within the same schooling group. An "experience" effect and a "knowledge obsolescence" impact are two competing effects that can be at work. Older workers typically have more job experience, making them more qualified at the same educational level, which causes the first effect. Older workers should therefore be less susceptible to technological unemployment because their duties should be more difficult than those of younger workers. According to the second effect, younger people tend to be more technologically aware than older people. For instance, compared to their 55–64 year-old peers with a same level of education, 25–34 year-olds tend to be more at ease using computers and mobile technology. For the best educated workers, who are most likely to work as data scientists, software developers, robot engineers, etc., this effect makes the most sense.

Mr. WGP Pinimal

Level 01
Department of
Industrial Management,
University of Kelaniya.

InfoStudio.

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InfoStudio, an informative tech-talk series featuring dynamic individuals from well-known local and international corporations, is another product of the Industrial Management Science Students' Association (IMSSA) of the Department of Industrial Management. InfoStudio conducts sessions for all Sri Lankan undergraduates to improve their capabilities, while providing insights into the information technology and management disciplines. These valuable and resourceful sessions cover topics related to management, IT, entrepreneurship, and soft skill development.The sessions are conducted via online platforms such as Zoom, the InfoStudio facebook page, and the IMSSA UOK YouTube channel.

Mr. Rathnapriya Wickramasinghe

InfoStudio

Mr. Malinda Alahakoon

The first session of the series for the year 2022 was conducted by Mr. Rathnapriya Wickramasinghe, the Deputy Director (ICT) at the Department of Census and Statistics. The theme of "System change in professional life by being converted to an evidence-based decision maker" was approached during the session.

The second session was conducted on the topic "The future of Sri Lanka and the role of youth from the perspective of economic development" by Mr. Dhananath Fernando, Chief Operating Officer at Advocata Institute.

Mr. Dhananath Fernando

The third session was conducted by Mr. Malinda Alahakoon, a YouTuber, who hosts the TechTrackShow. This session was based on "Online entrepreneurship and recipe of success."

The final session, to educate the undergraduates about the Google Summer of Code, was conducted by Mr. Eranda Sooriyabandara.

The sessions thus conducted under InfoStudio demonstrated its ability to mold young undergraduates to set up their mindsets when stepping into the corporate world. Each step taken towards the goal of InfoStudio, equips undergraduates with the skills, knowledge, and capabilities needed to succeed in the corporate sector.

Mr. Eranda Sooriyabandara

Scan for InfoStudio/

IM Sp

In 2018, I enrolled in Bachelor's in Management and Information Technology degree program at the Department of Industrial Management with the aim of grooming myself as an outstanding professional.

Within the last four years, the learning environment, along with the supportive and enthusiastic culture prevailing in the department, has resulted in my achieving the projected milestones in my academic journey. The lecturers always motivated us to nurture our character and attitude, sculptured with professionalism. The induction gained through the former MITians was invaluable since they always guided us and supported us when choosing the learning paths to develop our career goals step by step. During the COVID pandemic period, we were able to resolve all the obstacles we underwent with the methodical teaching mechanisms followed by the lecturers and complete our academic coursework related to our bachelor's up to the intended timeline. Also, the department did not hesitate to facilitate further new learning opportunities via renowned platforms like Coursera, edX and Udemy. I believe the inheritance received from the supportive panel of lecturers along with the challenging learning experience acquired during this pandemic period led me to drive my passion towards the Data Science Engineering field, ending up reserving myself as a Data Engineer in one of the renowned software companies in Sri Lanka.

The Department of Industrial Management is not only a dedicated institution for academics, but also comprises many programs designed for undergraduates to enhance their skills related to management and leadership aspects through the Industrial Management Science Students' Association (IMSSA). The undergraduates receive the opportunity to organize multiple events together with the industrial entities. Hence, the MITians receive the capability to showcase their maximum potential by interacting with the industry. Being the Chief Coordinator of hackX 2019, myself and hackX team were able to revamp the traditional hackathon culture, and conduct the event as a startup incubation program, recognized by many IT entities in Sri Lanka.

I strongly believe the inheritance acquired through this wonderful institution was the key factor to grooming myself as a confident professional, and the inspiration would definitely result in sustaining my personal and professional life in the future. I am forever grateful for being a part of the IM family.

Chiran Hewawitharana
Engineer - Data Engineering
Sysco Labs

pirits.

Blooming with Grace

I stepped into the University of Kelaniya as an MIT undergraduate in 2018 and completed my bachelor's degree in 2022, specializing in operations and supply chain management. Unfortunately, due to the COVID pandemic that started in 2020, our 3rd and 4th academic years were held remotely. We had to complete our individual development projects and research work during this period, which was really challenging. But it was a great and very valuable experience for us. During our undergraduate period, we got the opportunity to work with a very friendly and supportive academic staff who guided us not only in our academics but also in every obstacle we faced in our lives. They were always behind us to lend their helping hands; this supportiveness allowed us to successfully complete our academics and engage in extracurricular activities as well.

As MITians we get the opportunity to engage in a 6-month industrial training where we get a great industrial experience supervised through our lecturers as well. This exposure and the combination of management and IT knowledge and skills we get throughout our undergraduate period makes us unique in the industry and helps us a lot to secure very good career placements, entering the industry as fresh graduates. As I was a part of the university chess women's team throughout my undergraduate period, it helped me with soft skill development and analytical skills as well. I think these helped me to secure a satisfactory career placement soon after completing my final semester examinations.

The self-development, skills, competencies, and guidance we get from the Department of Industrial Management throughout our undergraduate period makes us successful individuals. It was a great opportunity to be a part of the IM family.

Anuradha Jayatilaka
Junior Procurement Executive
IPD Colombo

My decision to pursue a bachelor's degree in Industrial Management was entirely coincidental. I stepped into a whole new environment without having any clue about what was going to happen. Lucky me! DIM got me under her umbrella and nurtured me to be the person I am today.

The four years I spent at DIM were one hell of a roller coaster ride. From start to finish, it was all a learning experience for me. The academic staff, non-academic staff, seniors, and juniors were the perfect brand for ourselves.

DIM was full of happy moments. We had a variety of functions for all levels of study. Regardless of seniority, everyone enjoyed them to the fullest while tightening the bond between each other. Even today, we often interact with our seniors and juniors, which keeps the IM family alive.

I work in the ERP industry as a Systems Consultant. As a part of my role, I have to understand the manufacturing processes, optimize them and at the same time interact with ERP systems to facilitate the business processes. I still wonder how I would have managed this role if it were not for the modules offered to me in this degree program. I can vouch for anyone who is willing to take part in this program; this is one of the best places to get industry-ready knowledge and experiences that will guarantee a place in getting your dream job.

This request is for the youngsters joining DIM. University is not just a place to learn academics, but also a place to explore life. After admitting to this program, try doing extra-curricular work as much as possible while completing the academics. The staff will always be there to support you by any means necessary. Get the most out use of the resources provided by this beautiful place and use them for the betterment of the people who pay for your free education. That is what the DIM wishes for and what it stands for. Long live the DIM!

Lahiru Senarath Adikari
Consultant Associate
Fortude

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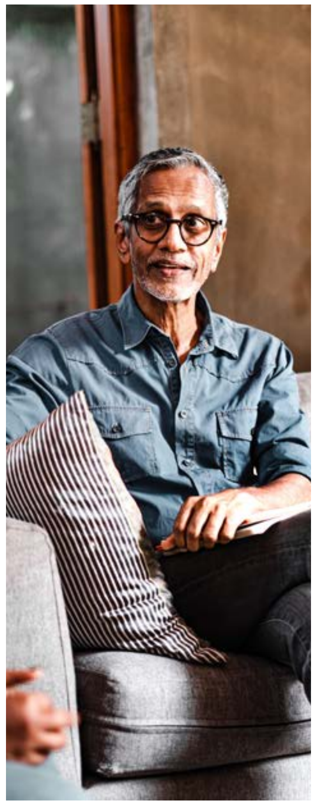
INTERVIEW

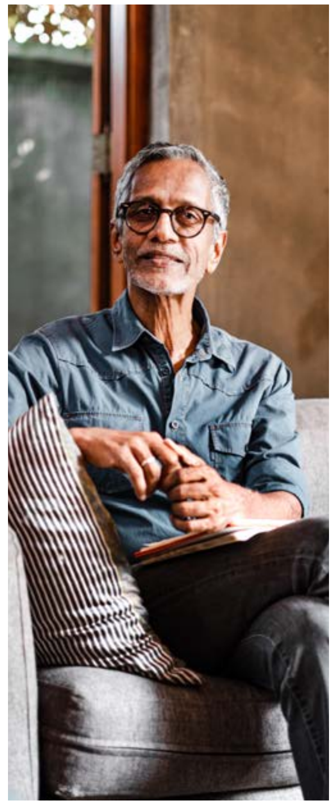
You cannot give up. Be resilient, fall, and rise.

Peter
D'ALMEIDA

Forging new paths of insight and creating spaces to foster cooperative brilliance are influential in one's personal and professional lives. An exceptional personality is able to relate authentic stories from that influence at every stage, from theater to business to life in general. Mr. Peter D'Almeida, Founder and former CEO / Managing Director of N*able Pvt. Ltd., is an idol who has taken his career on a varied trajectory as an entrepreneur, a leader, an actor, and a director.

Exposition Issue 18 proudly presents this prime personality, Mr. Peter D'Almeida, to apprise us of his story.





Q: In your own words, what makes Peter D'Almeida the person he is today?

I would prefer if others described who I am, but if you ask me what makes me who I am today, I will say that firstly, I am very thankful that I am socially conscious and concerned about the betterment of society. I am also happy that I have retained a sense of humility with all that has gone on in my life. I feel that if I am humble, I can learn more; than if I think of myself as a know-it-all. I am very thankful that I could retain a sense of humility.

The other is that I have a sense of purpose, which helps me stay focused. Even today, my sense of purpose is to do something to pull the country out of the current crisis.

Also, I think I can connect with people of different ages. I can talk to young children and people much older than me, and those of different genders. I'm happy that I'm able to connect.

But the things that I'm working on and can improve on, are being more mindful and calmer. And I think I have to learn to let go even more. I have let go a lot, but I can let go a lot more.

Finally, I am happy that I can still love people, things, and nature. I can love more than hate, which is a blessing. And I'm thankful for where I am today.

Q: How did your childhood and upbringing influence you both in terms of career and personality? And was there any particular incident that acted as a turning point?

I think people might misunderstand when I say this, but I'll explain why I say what I am saying; I think the turning point, although I didn't know it then, was when my father passed away when I was four years old, which completely changed our lives. We were a big family and had no income. I learned a few important lessons from that.

One is that we have to manage with limited resources. When you grow up with limited resources, you become sharper about the value of resources.

Secondly, we learned that people must come together to keep a family or a community going. My elder brothers and sisters had to forego their educational aspirations and go for work to maintain the family. When I first went to work, my salary came in an envelope - Rs.465. I was a bank clerk. We used to combine all our money and give it to our mother. The next day, we get the bus fare and the lunch money. Pooling your resources for the common good of the family and community is something I learned very quickly.

But most importantly, a result of losing my father was that I developed a very open mind. He was a conservative Catholic. He had very rigid views about what was and was not good. He would have conditioned me, and I would have become very narrow in my thinking. I became open-minded because he was not there, and because my mother was too busy running the family to interfere with my thinking. For those reasons - and don't misunderstand when I say this- my father passing away was a turning point in my life. Otherwise, I would not have been as free or as open-minded as I am today.

Q: Looking back, what was one incident that made a pivotal point in your career?

A pivotal moment in what led to the qualities that I have to be successful in my career came when at 15 years, I met this

veteran trade unionist, Marxist, socialist, and humanist Bala Tampoe. He was so well-read: from Marx to Casanova. I got exposed to Marxism, socialism, the history of revolutions, psychology, and philosophy. At 15 years, exposure to that kind of unique experience and vast knowledge made me who I am.

If I have to define a pivotal moment, it was me meeting him and absorbing all that he had to offer. And I think I would be half the man I am today if I did not have the encounter with Bala Tampoe and picked up all these qualities that led to where I am today, both in my career and personal life.

Q: You have worked at many institutions over the years. How have those experiences influenced you during the start of N*able?

One thing I learned is that in any situation, look for opportunities. People ask me about challenges, and I do not want to sound arrogant; it is not like there have been no challenges, but I trained myself to look for opportunities in every challenge. That is what I brought to N*able. We started with 12 people and were up against a few large and well established IT companies, including MillenniumIT, from which I came. People said we would not last six months, but our team said, "No, we can." Because we saw new and exciting opportunities that we could go after.

Also, over the years, I have learned how to pull together a team of people and how to convince them that there is a purpose and a plan. I learned how to tell them a story and bring them together. I learned how to lead them by helping them lead themselves.

I think those two qualities of looking for opportunities and pulling together a team are what I brought to N*able. Because this is not about you, it's about your team. You are only as good as your team. To put together a team and let them grow, not spoon-feed them or micromanage them, but give them a chance to look at the opportunities and then support them. I think those are the essential things I picked up as I started and grew N*able into a successful company.

Q: What challenges did you encounter on your path to an executive role? How did you overcome those challenges?

I never set out to become an executive; I did not have that in my head. But I think of all my roles as forms of leadership, even when I was a clerk at State Bank. I always wanted to lead myself. I did not depend on managers, as a result, whatever job I did, I found that I was leading myself and somehow leading others. I did not face any challenges as such.

It's just that I would take on or assume a leadership position in any role I was in. And when I say leadership, it is about leading myself and then being able to step up and lead others. I did not wait for someone to tell me or offer me a hierarchical position.

Q: You are renowned for your roles in the Sri Lankan media and entertainment industry. In which ways has that industry molded who you are as a person?

I am not sure whether the industry has molded me as such. But what I draw from being in that world is storytelling.

Storytelling is crucial. To form a company, I have to tell a story. To hire people, I have to tell a story. I have to tell a story to our customers and our partners. When you are acting, you are telling a story. You are playing a character and communicating. This is what I find common in my limited acting career and my professional and personal lives.

Most importantly, when you do a film, for example, you are perpetually reloading. I can look back critically on what I have done. I can always rewind and wonder whether it was good. Even in this last television series I did, Koombiyo, although my role was widely lauded, I can go back, look at it, learn from it and see what I could have done better. There's also audience feedback. Learning to be self-critical and not to sit back on your laurels, even though others may have thought that you were very good or excellent, I can see how I could have done it better. That's an important quality I learned from being in that industry.

Q: In your opinion, what skills do prospective employees lack that are most sought by employers?

Some years ago, PricewaterhouseCooper did a study. They interviewed 3000 CEOs across the world. And they came up with what they identified as pivotal talents: creativity, problem-solving, collaboration, and communication. But underlying all this is curiosity. If you are deeply curious, there is nothing that you cannot discover.

One of the problems we are having is that people cannot solve problems. Once, at N*able we wanted to hire people for our predictive analytics practice. We interviewed three students who topped their batch at the university in statistics. Sadly, we could not hire any one of them. They knew their subject but simply couldn't apply their learning to solve problems: You give them a problem and ask them to use regression, they will use it well. But they cannot look at a challenge and think, "You know what, I could apply regression to address this problem." That is what we found lacking. Our education system does not allow you to solve problems because, to solve problems, we have to fail. In our system, failure means match over, and this is the problem. If you get five answers right, out of ten correct, the student who got five is not celebrated. No one says, "Oh my God... we gave you ten problems, you solved five, let's solve another five." We cannot promote problem-solving and curiosity when our education system is not geared to encourage them. Our system is entirely based on memorizing and reproducing. This does not encourage problem-solving.

Further, it does not encourage collaboration either. Students are made to work alone. Today in the working environment, you can't shine as a star by yourself. Individual stars can sometimes be toxic. You would be great alone, but if you can't collaborate, we will be reluctant to keep you on our team. We want people who can work together.

These are the qualities that we find important. But above all, curiosity. With the internet, if you are curious, the whole world is open to you.

Q: As an established businessman, what words of advice would you like to give budding entrepreneurs to ensure they are on the right track?

Entrepreneurs must be very clear about the business they are trying to create. They must have a well-defined purpose. They must be driven by that purpose. In other words, you should be able to say, "I am either solving this problem or I am creating this new experience." I have learned from my early days in marketing that people buy only one of two things. They buy solutions to problems or they buy good feelings.

You buy a pen to write, but you buy a Montblanc or a Parker because you want that experience. An entrepreneur should be very clear: I am going to offer a new, a cheaper or a better solution, or else I am going to create a new and exciting experience.

The second important thing I have learned, and am still learning, and something I could have done better, is to have clear measurements of success. You have a purpose, you know what you are going to build, but you must have clear measurements of how you are progressing. I think very few hopeful entrepreneurs define what those moments are. Most people use profits as the measurement, but I say no. Look at the steps along the way. Profits are just an outcome. When Google set out, they did not say they wanted an algorithm that would take 10 milliseconds, they said they wanted it instantaneously. So, they pushed themselves to answer instantaneously. To get there, they had very specific measurements. I think that is what people are lacking. To be successful in building any business, you must be able to see opportunities in any situation. Spotting opportunities is what makes a great entrepreneur. Finally, nothing tops hard work. You must work very hard to get there.

Q: You value creativity, curiosity, and critical thinking. Do you have any suggestions to develop these traits in an entrepreneur?

This has been a problem, and I have often thought about whether you can teach people how to be creative and naturally curious. To be honest, I am not too sure. As a child, you are very curious and have so many questions, and you are told not to be too smart! Adults put so many restrictive layers on children that I sometimes wonder whether you can unravel these layers and get back to where you were as a child, naturally curious and creative. But perhaps we can try. Let me give you an example. Recently, I went to a less-privileged school in Maawala attended by very poor children. Some university alumni have put up a fund to provide them with one meal a day. And they cook in the school. I got the Grade 6 class. I spoke to them for a little while. I told them my childhood stories. Then I asked them to work in groups to come up with one challenge they could overcome or something they recognized the school could do better. After a while of discussing, they came up with so many ideas. The principal was shocked. The school or their parents have never told them to think of ideas and problems. That is what happens in the university system too. All the lecturers and academia tell you what to do, instead of letting you develop the curriculum. I think when we put people into groups and ask them to come up with ideas, they will. Without supervision, grading, and marks, I am sure they will come up with ideas. They will pull things from their past creativity and come up with alternatives, ideas, and possibilities, which is what creativity is all about. I think there is hope.

Q: From your perspective, what should Sri Lanka prioritize to bring more foreign currency into the country?

Sri Lanka's problem today is not simply about bringing in foreign revenue. We are losing people as we speak. Do not forget that we are bankrupt. No money to pay debts, borrow or generate. We are losing hundreds of thousands of our best resources. When you are bankrupt, you require good resources to build back. As long as we have this type of corrupt system of government, and if we are indifferent, nothing will change. People will not come back. Why would they, as long as the leaders who brought the country to bankruptcy are still in power? The IT industry can help generate foreign currency but we are rapidly losing people. A group of IT professionals has come together to address this and some of the larger problems faced by people. We recognize that what we need is a system change. If we can make some fundamental changes, then we create the environment for people to stay and do what they are best at doing. Then the foreign revenue will come.

Q: If you could write down a recipe for success, what would it be?

How do you define success? That is the point. I think too many people are looking at success as material success like money, or having a home or a car. Fresh graduates out of university think they should get a job with a high salary, and within two years they should get a car. They get into debts and loans to build a house. Some even get into debt just to have a grand wedding reception. If you define success with that kind of criteria, then I have no recipe for success.

If you define success as an accomplishment of a purpose or an aim, then it differs. There's no recipe for success if you have not identified what success means to you. Today for me, for example, success will be if I can mobilize people to resolve some of the many problems we face. This is what I am doing with the IT professionals in the country. In this instance, we define success as achieving some of our specific and measurable tasks around a few well-defined objectives. I think people should define what they see as success and they have to write their own recipes. That is not something to be prescribed. Identify a purpose or an aim and determine what it looks like if you reach that aim or purpose; then you can go back and write your recipe. You learn the recipe yourself. You try, learn, and fail. Perseverance is the only recipe for success. You cannot give up. Be resilient; fall and rise. Relentlessly pursue your purpose or your aim.

HOW YOU DID IT?

"How you did it?" is an episodic session series that provides a platform for undergraduates to be inspired by the success of others. It is an opportunity to learn from others' experiences and thrive in their future careers. The session series was conducted for the third consecutive year and it will be conducted throughout the year.

Episode 06 - Management or IT?

Conducted by

Mr. Ahmed Ifhaam
Senior Engineer
Nagarro

Ms. Shilpa Gunawardana
Customer Service Manager
Modern Trade and E - Commerce
Unilever

How you did it? - Episode 06 is a sequel, designed to guide and enlighten MIT undergraduates to reach their goals by helping them to distinguish what the Management and IT sectors have to offer. The sessions were conducted by two proud products of the Department of Industrial Management.

Episode 06

Episode 07 - Professionalism on Social Media

Conducted by

Ms. Hasulie Dias Abeysinghe
Assistant Manager
Corporate Communications
Timex Garments (Pvt) LTD

How you did it? - Episode 07 was aimed at guiding the undergraduates on how to maintain their social media presence in a professional manner. It also brought out the importance of maintaining a social media presence for undergraduates as well as budding entrepreneurs.

Episode 07

Power Skills to Stand Out as a Data Scientist

By **Nelumi Opatha** | Level 04

"Knowledge is power"

This is a famous proverb by the British philosopher Sir Francis Bacon from his publication Meditationes Sacrae, which dates back to 1597. It has been repeated and expanded by many throughout history. This idea is often brought into discussion, especially in the business world, as well applied knowledge is a competitive advantage that creates power.

Wisdom hierarchy

Moving backward to see how knowledge is created, the wisdom hierarchy illustrates that knowledge is defined in terms of information. Information is defined in terms of data. Data refers to unprocessed facts and figures. However, data alone can tell you nothing. Transforming data into information is the fundamental objective. The information in action can be defined as knowledge, and the purpose of it is to be useful in a particular domain or different domains. As you go forward reading, you can now figure out that knowledge can be derived from data that empowers people, businesses, and almost all the things around us. Therefore, let us refine the proverb into, "Data is power".

What is data science?

The field of study known as data science works with massive volumes of data using cutting-edge tools and methods to uncover hidden patterns, derive actionable insights, and make strategic decisions.

A role of a data scientist

A data scientist is someone who analyzes business data to extract actionable insights. He follows a series of steps to solve critical business problems. As the first step, a data scientist attempts to understand the context of the problem and the problem precisely by asking the right questions from stakeholders. Then he determines the required sets of data and the correct set of variables.

As the next step, the data scientist gathers structured and unstructured data from multiple sources. After the data is collected, he transforms the raw data into a format that can be used for analysis. This process involves data cleaning and validation, which ensures consistency, comprehensiveness, and accuracy. Then the data is fed into the analytical system, and he analyzes and identifies patterns and trends in data. From that, the data scientist interprets the data in a way that helps to resolve the existing problems. Finally, he prepares the findings and insights to share with the business stakeholders and communicates them to relevant parties. Does it sound complicated?

Maybe, but here is one major reason why you should aspire toward a career in the data science domain. As Glassdoor and Forbes suggest, the demand for data scientists will rise by

26 percent by 2026. Therefore, it is worth putting effort into developing both technical and professional skills in you to fit into a job role in the data science domain. Keep reading to get to know some valuable power skills that will help you outshine the rest!

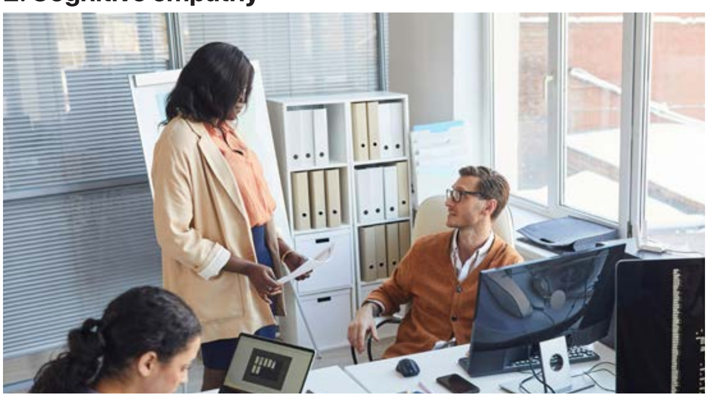
Power skills

In its 2022 Workplace Learning Trends Report, Udemy Business introduced the term "power skills" to characterize those attributes once referred to as "soft skills," elevating them as the top competencies required to succeed at any level within an organization. The term "soft" skills does not stress the importance of mastering these skills; therefore, "power skills" sounds a better fit. Here are some significant power skills that are not common in discussions but are very important in today's dynamic working environment.

1. Confidently protecting your turf

When it comes to your job role and job description as a data scientist, not everyone will define it the same way. The position described to you at the interview, and your role at work can be completely different from each other. This can happen in a business context where the management is struggling with implementing a sound analytics strategy. In such a situation, as a confident data scientist, you should proactively suggest the kinds of projects you can get involved in that benefit the team and the whole organization. Select the projects which are challenging as well as rewarding. Showcase your analytical skills and demonstrate the best version of the data scientist in you.

2. Cognitive empathy



As a data scientist, there are times that you have to put yourself in someone's shoes to figure out the way they think. With cognitive empathy, we try to understand someone's thought process that led to a particular decision. When observing a dataset, there can be certain things that you notice but which are not visible in the business context. Also, there can be things that are not visible in the data set but can be seen in the business context. Therefore, having cognitive empathy supports interpretation and generation of insights and, finally, to build better models.

3. Curiosity

A data scientist needs to be curious to find answers to known business problems and uncover overlooked solutions. The key to stand out is to continue asking questions, pushing the boundaries of your knowledge and keep exploring things. Being curious in these situations is beneficial. However, you should know when to stop exploring and start taking action with the data you possess. Keep in mind that there is always something to learn, regardless of the type of data you are working with.

4. Persuasion

Do not expect persuasion to be easy. After you complete building your solution to the business problem, you should convince the stakeholders that you have solved their problem. These stakeholders may be skeptical and non-technical, but you have to present your solution in a way that everyone can grasp it. The teams will be questioning you, doubting the numbers you show and asking to add more features to the model, however you should have the ability to defend your solution both logically and practically. Before presenting your solution, ask yourself, "if I were them, what would I be focusing on?"

5. Commitment to hone your craft

As a data scientist, you should know the new tools, technologies, and trends out there. Reading related books, reading online content, and attending conferences are a few things you can do to invest your time. You should ensure that you keep up with the pace of technology and develop professionally to fit into the ever-changing technology. The technical skills you have now could be completely obsolete in a few years or maybe in a few months. Therefore, being up-to date is crucial to add value to your position.

You can focus on these power skills to outshine yourself as an inspiring data scientist in the organization. Some of the skills you just read might not have been in your discussions, but many of you might have felt the importance of these while playing the role of a data scientist at your organization. Having the required technical skills is necessary, but the power skills will make you stand out.

Ms. Nelumi Opatha

Level 04
Department of
Industrial Management,
University of Kelaniya.

INTER SCHOOL HACKATHON 5.0

GIVE SHAPES TO IDEAS

hackX Junior 5.0 is the 5th resplendent iteration of the inter-school hackathon organized by the Industrial Management Science Students' Association, Faculty of Science, University of Kelaniya. Backed by the theme of "Give Shapes to Ideas," the competition provides a platform for the bright, young minds of school children to unleash their potential and transform their ideas into reality. With the approval of the Ministry of Education, hackX Jr. 5.0 set the background for the school community to bring forth innovative ideas that may contribute to global knowledge and accelerate the local economy.

hackX Jr. 5.0 consisted of four phases: Awareness Session, Online Workshop Series, InnoX - the semi-finals and the Grand Finals. The Awareness Session was held on the 30th of July 2022 via Zoom and a Facebook live. An overwhelming count of over 300 participants joined the Awareness Session. They were briefed on the competition guidelines, application process and timeline. Ms. Nevindaree Premarathne, Senior Manager - Startup Ecosystem Development at ICTA, enlightened the participants on how to develop an entrepreneurial mindset. Afterwards, Mrs. Pulani Ranasinghe, the Founder and Director of Loons Lab, addressed the audience on idea generation and how to write the perfect project proposal. Three knowledge-sharing sessions were held parallel to the Awareness Session. Mr. Vartharajah Kumaraguru, the CEO of Youth Business Sri Lanka, conducted an Inspirational Session to motivate the youngsters. This was followed by a Level Up Session, conducted by Mr. Chamil Jeewantha, Software Architect at Multiplier. An informative session on Mobile Application Development was held by Mr. Sheshan Gamage, Technical Specialist at Wiley Global Technology (PVT) Ltd, on behalf of the SLASSCOM TechKids program.

From over 130 proposal submissions, 25 proposals were shortlisted to compete in the semi-finals. Two online workshops were held to guide the semi-finalists. Mr. Mafaz Ifharm, Co-founder and Chief Storyteller at Show & Tell, conducted a workshop on How to Pitch your Idea. At the second workshop, Mr. Dimuthu Ponnamperuma, Chief Strategy Officer at Accelozo, acknowledged the participants on Digital Marketing and Growth Hacking.

InnoX, the semi-finals of hackX Jr. 5.0, was held on the 11th of September 2022 via the Zoom platform. The teams got the opportunity to pitch their innovative ideas before a distinguished panel of judges, composed of Mr. Vartharajah Kumaraguru, Ms. Ishanka Rathnayaka, and Mr. Buddhika Jayawardena. Based on the ingenuity of the idea, its feasibility and potential, 12 teams were selected to compete at the Grand Finals.

Finalists were tasked with developing their idea to a prototype level for the Grand Finals. Accordingly, Mr. Migara Amithodhana, CEO and Co-founder of Magicbit (PVT) Ltd, guided the students on Most Viable Prototype Development. The final workshop of the series was conducted by Mr. Mohammed Fawaz, Founder and CEO of CurveUp, on the Lean Canvas model.

The Grand Finals of hackX Jr. 5.0 was held with much grandeur on the 12th of October 2022 at Ideamart, the auditorium of Dialog Axiata. Preceded by a few encouraging words from Professor

Janaka Wijayanayake, Head of the Department of Industrial Management, the finalists took the stage. They presented before the judges a working prototype of their idea, along with their business model and future implementations. After a tough competition, Team CodeGMV of Gurulugomi Maha Vidyalaya, Kalutara, emerged champions of hackX Jr. 5.0. Team Edu-Me from Hartley College, Jaffna and Team DetectInnovat of Zahira College, Colombo, emerged as the first and second runners-up respectively. Dr. Dilani Wickramaarachchi, Ms. Shiroma Dahanayaka, Ms. Nevindaree Premarathne, Mrs. Pulani Ranasinghe and Mr. Dushantha Ranwala comprised the honourable panel of judges at the Grand Finals. Corporate sponsor representatives, parents, teachers, and undergraduates also added color to the event.

An honourable mention should go to the official partners of hackX Junior 5.0, whose contributions led to the success of the event: Official Silver Partner Informatics Institute of Technology, Official Bronze Partner Java Institute for Advanced Technology, Official Venue Partner Ideamart, Official Food Partner Prima Kottu Mee, National Partner Information and Communication Technology Agency of Sri Lanka, Official Strategic Partner SLASSCOM TechKids, and the Official Knowledge Partner Loons Lab.

Thisara Dilshan and Hiruni Tennakoon from Level 2 were the chief coordinators of the event, and Team hackX Jr. 5.0 consisted of Ashen Shanuka, Harini Jayasundara and Yeshan Wickramasinghe from Level 2. The event would not have been a success without the immense support extended by the lecturers, academic and non-academic staff members, and undergraduates of the Department of Industrial Management.

www.hackxjr.lk

INCO

INVENTION CONVENTION

For Engineering Smart Systems for Industry

The InCo competition is a one-of-a-kind event that fosters creativity and encourages participants to think outside the box. It serves as a platform for nurturing brilliant concepts and sparking entrepreneurial ideas among the undergraduates at the DIM. It is an opportunity for participants to bring their innovative ideas to life, and present them to a community of like-minded innovators and industry experts. The competition is open to all second-year undergraduates who follow Management and Information Technology and evaluations are based on the innovation and marketing of the product or service.

InCo 2022, organized by the Department of Industrial Management, Faculty of Science, University of Kelaniya, was held on the 30th of November 2022, at the department premises. Academic and non-academic staff members and industry representatives adjudicated, made suggestions, and shared their industry experiences with the competitors.

After a fair evaluation by a prestigious panel of judges, the winners were revealed. Winning teams fell under the categories of "Innovation," "Marketing," and "Best Innovative Idea."

Team A-Listers won the Overall Championship, emerging second runner-up in marketing and first runner-up in innovation categories, with "Aurora," a clothing brand of natural-colored clothes. The team consisted of Sajani Gandhira, Navodya Pasindu, Sathsarani Silva, Kenneth Pinto, and Nimesha Aththanayake.

Team Producera, consisting of Muditha Samarasinghe, Isal Laksika, Amaya Senarathne, Harini Jayasundara, and Thilini Bhagya won the marketing category and became the second runner-up in the innovation category, with their product "Pera Munch," a guava-based nutri bar.

Team Firestorm, consisting of Manod de Silva, Naveen Jayathilaka, Nimesh Heshan, Kavisha Amarasinghe, and Hiruni Hathnagoda, won the innovative category with their product "Kiri Podda," a domestic coconut milk extraction machine.

Team Ground Breakers, emerged as the first runner-up in marketing with their product "CookPal," a meal kit business. Raduni de Abrew, Dinithi Samarathunga, Nihim Abhayarathne, Tharindu Weerathunge, and Sarith Hasarinda were members of Team Ground Breakers.

Team A-Listers

Overall Champion
Innovation : First Runner-up
Marketing : Second Runner-up

Team Producera

Marketing : Champion
Innovation : Second Runner-up

Team Firestorm

Innovation : Champion

Team Ground Breakers

Marketing : First Runner-up

The Stories of Undergraduates
from their Perspectives

365 x 4 | **One year. 4 levels. 4 stories.**

Embarking on a life-bound journey together, each with their own dreams and aspirations.
But despite their individual goals, they are all united as the IM family.

The beginning of the journey

The Freshman.

Q: What was it like for you to go from a school student to an undergraduate at the Department of Industrial Management?

Being selected to follow MIT at the Department of Industrial Management is the best thing that has happened to me. Of course, being an undergraduate is different from a school student, but being an MITian in University of Kelaniya is a rare, lifetime opportunity. Even though absorbing the difference and grabbing the new environment was not quite challenging, I think the department handled us as pros as always. After coming to the department, I started to see the world from a bigger perspective and imagine myself in a very new way that I've not seen so far. It made me dream and wonder about me being a whole new, different person that I've not seen in my school time. Most importantly, with DIM, I believe that the best is yet to come.

365 x 1 | **One year. 4 levels. 4 stories.**

Q: How do you think this degree program will help you pursue your passions?

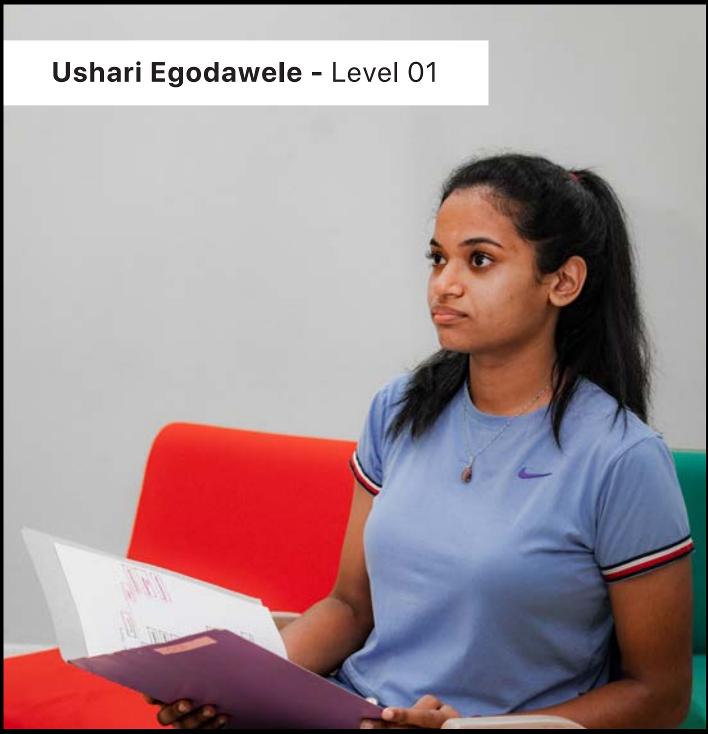
Learning management and IT even though you are a science student is not an easy chance. I was passionate about starting my own business, but I didn't know how to pursue that with my study area. Except after starting to study at DIM, I gradually started on drawing and sketching the plan. The content of the degree program allows us to think as more analytical and logical problem solvers with the ability to handle big data. I still have a long way to go, but I am confident that I can improve further and be a dream chaser.

Q: According to you, what inspires the MITians the most to become outstanding?

They are guiding us to imagine, think novelly, and be innovative, and encouraging us to go beyond our comfort zones while learning who we really are. I think these concepts are breaking barriers in our heads that we had in our school period, and it's amazing how they are guiding us step by step. They encourage us to get involved in a variety of extracurricular activities so that we can become well-rounded members of society. Moreover, the academic staff, the non-academic staff, and the seniors are supportive and motivated to carry us with them without any difference. This culture in the department itself is the main difference I see from other departments or universities in our country. That, in my opinion, is what propels the department to the next level.

Q: After entering the Department of Industrial Management, what changes have you already noticed in yourself?

I started to see the world differently. I learned that diverse talents can be developed in a person and can bring about significant change. At first and still, even the subject contents were hard to follow because they were entirely new to most of the students. But it creates flexibility and the skill of learning quickly in students. I also discover more capabilities in myself, such as being good at communication, handling people, discovering their skills and abilities, and identifying the risks beforehand. Also, I should improve my skills in public speaking, communication, and leadership throughout the next few years of study to become the person I want to be. I'm hoping that DIM can help me gain these qualities.



Ushari Egodawele - Level 01

Settled into life at uni

The Sophomore.

Q: How has your undergraduate experience been so far?

As an undergraduate, I am a part of a diverse community of students from various backgrounds. I attended lectures, seminars, and tutorials where I learned about various subjects and gained valuable knowledge and skills. I also had the opportunity to participate in extracurricular activities, such as clubs and sports, which helped me develop my leadership, teamwork, and time-management skills. The simple truth is that these two years changed my life in a way I could never imagine.

Waruna Sri Wickramasinghe - Level 02

Q: What is your preferred area of study; Management or IT? And why?

Definitely the answer is IT. The study of Information Technology provides a foundation for a challenging and rewarding career, and the opportunity to make a real impact in the world through technology. Since our degree program delivers up-to-date content, with my social skills, if I were to complete my degree successfully, I would get a high- salary job for sure.

365 x 2 | **One year. 4 levels. 4 stories.**

Q: You were the Chief Coordinator of hackX 2022, the inter-university startup challenge. Can you share the experience you gained from organizing such a successful event?

Organizing hackX 2022 was a complex and challenging task that required strong leadership skills, attention to detail, and the ability to work well with the team. As the Chief Coordinator, I oversaw all aspects of the event, from securing funding and recruiting participants to coordinating with the hackX team and ensuring that the event runs smoothly on the day of.

Q: As an MITian who also represents the university hockey team, how would you put teamwork and dedication into your own words?

Teamwork and dedication are crucial qualities for success. Teamwork involves working together with others towards a common goal, while dedication refers to a strong commitment and perseverance towards that task or goal. Together, they create a powerful combination that enables individuals and teams to achieve great things by combining their strengths and maintaining a focused and determined effort.

In my own words, teamwork and dedication are the foundation for success in hockey and in life. By working together and staying committed to our goals, we can achieve great things and make a positive impact both on and off the field.

Q: How do you strike a balance between your studies and extracurricular activities?

Balancing studies and extracurricular activities as an undergraduate is simply prioritizing responsibilities, setting achievable goals, and using effective time management strategies. Making a schedule, breaking down tasks, and knowing what is most important helped me to allocate time effectively and maintain a healthy, well-rounded lifestyle.

Q: What improvements have you observed in yourself since becoming an undergraduate of the Department of Industrial Management?

Before these two years I was a student who enjoyed his freedom to the fullest, but since the university started I would say I have become a more organized and disciplined person. During my school days I was more theory-oriented but now, I would say I'm a very practical guy. All that credit goes to the department without a doubt. Other than that, the department has been the bed rock for the improvement of my analytical and problem solving skills.

Q: In the next academic year, current level 2 undergraduates in the department will be seeking internships. Hence, what will be your preferred field for an internship?

Right now I'm following the software engineering path in my degree and my plan is to become a software engineer one day. Since I would like to gain hands-on experience in that field and start my journey there, I would like to start my internship as an intern software engineer in a reputed company with international exposure.

Q: Finally, can you express what it is like to be around your fellow MITians?

Being around my fellow MITians is a fulfilling and enriching experience, offering opportunities for relationship building, personal and professional growth, and exposure to diverse perspectives and experiences. It creates a supportive and collaborative atmosphere that fosters learning and development. Being a part of the IM family is the highlight of my university life.

L3

A glimpse into a new world

The Intern.

Q: What is your chosen career path and what ignited that passion?

As MITians, most of our career paths would be in the IT industry since we have the privilege to excel in both IT and management domains. With that, my passion is in Business Analysis and Project Management, where I get to deal with different stakeholders at different levels across the industry. I've always been someone who enjoyed being with people and hanging around with them while tackling new challenges each day. I think that's one of the main reasons that I chose Business Analysis as a career path over others.

365 x 3 | **One year. 4 levels. 4 stories.**

Q: How crucial do you think it is for undergraduates to participate in extracurricular activities throughout their time at university?

My answer to question would be, "That's probably the best way you can experience the so-called University Life," since participating in extracurricular activities is crucial and important in an undergraduate's life. You cannot even think of getting the whole package of experiences unless you are participating in any of the extracurricular activities. There are many sports, clubs, and societies, as well as student communities and other events that undergraduates can get involved in. It provides undergraduates the opportunity to develop new skills, identify strengths and weaknesses, and help them grow their network. Each of these will be a whole new experience for them. Thus, it is really important to participate in extracurricular activities, while balancing academics to get the maximum out of your university experience.

Q: What inspired you to participate in and win several hackathons, and what kind of support did you receive from the department to accomplish these victories?

Actually, it was a whole new experience for all of us on our team. Our team leader had many creative and innovative ideas, and we as a team thought of applying to different competitions, just for the sake of getting new experiences. But along the journey, we realized that our ideas and the solutions we came

Dilshan Hettiarachchi- Level 03

up with were creative and competitive enough to meet the standards. And we are really grateful for the immense support that we received from the Department of Industrial Management and our fellow undergraduates. It would not have been possible for us to compete and achieve some victories if not for their support and guidance. And I truly believe that it has been a huge milestone in all our team members' lives.

Q: You were the former captain of the UOK Athletics team. How did you manage to balance your academics with extracurriculars?

I wouldn't lie and say that there hadn't been hard days because managing both takes a lot of effort and time. It's all about how brilliantly you allocate the right amount of time and priority to both. I don't treat sports and academics differently, and thereby I have always tried to balance them out equally. There is no hard and fast rule as to how you manage, it comes with practice and also dedication. Consistency is the key, with consistence and perseverance comes success in everything that you want to achieve.

Q: As an all-arounder, how has your time at university benefited you so far in honing your talents and abilities?

Honestly speaking, the time I spent at our university was extremely limited due to the never ending pandemic that struck the whole country. Consequently, we were all forced into online education and digital learning, which come with their own share of upsides and downsides, just like the two sides of a coin. But that didn't stop us from showcasing our abilities because we used virtual platforms to organize a variety of programs and events. That being said, the university provided us with multiple opportunities for public speaking, executing and leading programs, moderating/oration and they also expanded their scope to overseas programs that helped us work alongside international candidates and upgrade the standards of ourselves and the MIT program.

Q: How is your experience at SRIQ Corporation (Pvt) Ltd as an intern?

In one word, it's astounding. Every workplace is defined by the human resource and of course SRIQ has a lovely dynamic staff, that is always very welcoming and enthusiastic to guide us through this internship journey. Besides the professional side, we celebrate various events together and it's been only few months now, but I feel like I belong there already, like one place, one family. To speak on the work itself, as interns we don't have an obligation as to what we pursue, we are given the freedom to pick the cup of tea we like, and my designation is Functional Consultant and I really enjoy working there.

Q: Could you go into more detail regarding the exposure to the industry you are receiving through the internship?

Our internship is based on Microsoft Dynamics which is integrated with managing and providing ERP solutions for companies that look forward to migrating to Business Central. Microsoft Dynamics is one of the leading business solutions that companies seek in order to move from legacy systems to more advanced ones that are enveloped with extended Warehousing, Purchasing and Procurement, Sales and Marketing functionalities. Moreover, we learn the insights of Power BI and Power Apps and incorporate them into our learning to enhance the business process.

Q: In what ways has the MIT degree program aided you with your internship?

First things first, without MIT I wouldn't have found this job, so MIT took the initial step to place us in different companies and brought us into this corporate world. Next, the degree program itself is a very integral program that consists of various resourceful courses that help facilitate this internship session. Also, the experience we gain through the intensive learning process, practical sessions, presentation skills and technical knowledge, help us in each and every point of our internship. MIT did not spoon-feed us everything, they paved way for us to self-learn various aspects of the study, developed our analytical skills and this is a remarkable strategy for personal and professional growth.

Q: After graduating from the Department of Industrial Management, what do you envision for your future professional life?

I think we need to love what we do. Whatever we do with a sense of interest never goes out of style. And never stop learning no matter how high we climb up the career ladder or how much knowledge we gather, because life has its dramatic way of teaching us new things every day. Although time and tide don't matter in this learning phase, we need to be mindful that now is the time to establish a solid foundation for the benefit of our future. Further, without sticking solely to one particular field, having our hands dipped in every pie will provide us with vast acquaintance, knowledge and experience. It helps us expand our career opportunities without limits. Supply Chain Management has been my first professional choice but with the experience I gained at SRIQ in ERP Consulting and Microsoft Dynamics I would like to progress ahead in any one of those fields.

Q: As an MIT undergraduate, what message do you have to present to your fellow undergraduates?

There's a saying that the university is a kitchen where you get all the ingredients you want, and you are the chef who is responsible for preparing yourself the best meal in the whole world. At university, you will get different opportunities to participate in various events, competitions, clubs and societies, sports and so on which will help you to mold yourself to the person you want to become in the future. Therefore, my advice would be to grab as many opportunities as you can, that come along your journey and you will never regret that decision once you start getting satisfaction when you look back on yourself.

About to step into the industry

The Senior.

Q: How did your first three years at the Department of Industrial Management influence you to continue to the 4th academic year?

As undergraduates in the Department of Industrial Management, we engage in various activities apart from academic activities. Almost everyone engages in extracurricular activities as well as in organizing various events at the department. Therefore, we all get together and work as one team to make everything a success. It was a wonderful time we had in our first three years.

Moreover, the lecturers, non-academic staff, and my batchmates had a strong relationship as the IM family rather than an academic-oriented department. With that homely experience, I could develop both technical and soft skills in a relaxed mind while improving my professional competencies. That experience at the department influenced me to continue to the 4th academic year.


Ruwanthi Perera - Level 04

Q: What steps did you take to get to Level 4 undergraduate status in one of the most important degree programs, and what difficulties did you encounter while pursuing your education?

It was a wonderful time in my first year at the Department of Industrial Management since all the lectures were held at the university premises. At the same time, we had various events where we got together and worked as one team. I participated in some startup challenges and hackathons as well. It was a time full of learning and amusement at the same time.

365 x 4 | One year. 4 levels. 4 stories.

I used my free time to play chess at the University Chess club. But from half of my second year on, we had to face the COVID-19 pandemic and transform it into online lectures, and we were also engaged in organizing cooperative events. Therefore, changing to an online setup was a bit challenging initially because it was new to all of us.

However, I had to physically go to the office during my internship, where many of the people were affected by COVID-19. I had to come across that and perform my work there with all the challenges around the environment. After three years, I was eager to return to the department and continue on to the 4th academic year since it is the final year of my undergraduate life at the Department of Industrial Management. I always wanted to explore new things and grab knowledge while engaging in extracurricular activities. Hence, the final year is the best opportunity, where we have to conduct research according to scientific methodologies. It is a totally new experience that adds more value to academic activities and future careers.

Q: What is your specialized area in the degree program? And how did you get passionate about that area?

I specialize in Operations and Supply Chain Management. After my A/L examinations, I applied for B.Sc. in MIT with the ambition of choosing Operations and Supply Chain Management as my specialization area. During the first three years, I realized the impact the Supply Chain industry has made on nearly every facet of the modern lives of people. It is one of the most diversified fields in the business sector as it is interconnected with all other functions. Because supply chain management is rapidly expanding, and it is also expected to grow even in the height of a recession. There are various career path options, like different sub-sectors, where a person can learn a lot while ensuring a positive growth career with a good salary. Hence, during our degree program, my passion for following this specialization area was improved with the vast area of modules that we had, capturing both IT and management.

Q: How did the MIT degree program benefit you in achieving a successful internship?

I had the opportunity to work as an intern in one of the leading apparel exporting companies in Sri Lanka, which is also the largest apparel tech company in South Asia. I was working as a shipping and logistics intern while also engaging with some of their automation projects. It was an amazing experience. I saw the real-world implications of what we have learned over three years. I had an opportunity to learn the export and import process of apparel in Sri Lanka, and apply what I know to automate their daily routine activities, such as documentary work. Those six months were my very first industry experience, and they gave me an understanding of the corporate world and workplace culture, best practices, and many more. All my gratitude goes to my degree program because our department has very good coordination for providing internship opportunities for the whole batch.

Q: How did you land a Microsoft Learn Student Ambassador opportunity during your undergraduate period? In what way did MIT pave the way for this opportunity?

Anyone enrolled as a full-time undergraduate in an accredited academic institution can apply for Microsoft Learn Student Ambassador Program. MIT degree program was the foundation for me to apply and successfully get through the selection process. Because as an MIT undergraduate, I was passionate about exploring new knowledge and skills while sharing what I knew with others. To explain a bit about the Microsoft Learn Student Ambassador Program, it provides plenty of opportunities for you to learn and lead, allowing you to make a difference and empower those around you. Student Ambassadors get the opportunity to connect with other students worldwide, and their online communities, get training and certifications, and tackle practical challenges, all while building key technical and soft skills to help them succeed.

Q: How did your experience at the department pave the way for the success of your research project?

It was a completely new experience to conduct research in the fourth year of our degree program. Our department always encourages us and helps with every part to do it successfully. All the lecturers, including our research supervisors, support us a lot in carrying on the research projects without any problems. We have received immense knowledge from all the conducted sessions, especially data analysis workshops.

Q: How has the MIT degree program helped you lay the foundation for advancing to a full-time professional career?

MIT degree program paved the way to a professional career from the first day at the Department of Industrial Management. I gained knowledge in numerous avenues, and I improved my soft skills day by day. This degree program laid the foundation to build my professional image with more productivity and efficiency. How we learn, behave, talk, and work has been shaped throughout the years. Not only having a versatile knowledge of management and Information Technology, but we also develop our careers through personal progress development activities, outbound training, and many more.

The internship program was another turning point that gave me an excellent experience to start a professional career. Those six months I completed was a great achievement and a fresh beginning.

Q: As an ambitious student with a distinguished degree, in what ways have your experiences led you to become the aspired person you are today?

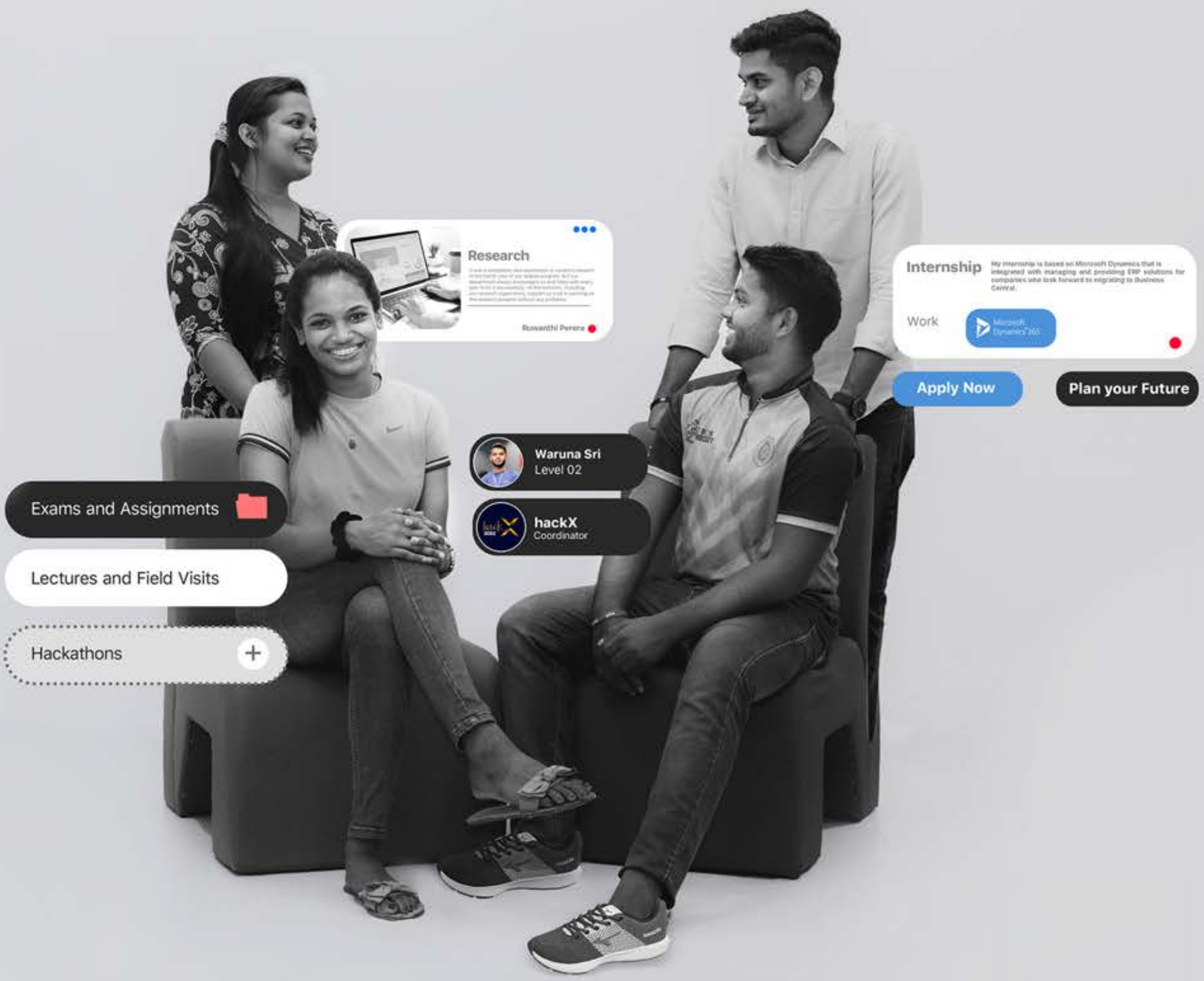
All these four years at the Department of Industrial Management were unforgettable roller-coaster rides with never-ending learning opportunities. Not only excelling in academic activities, the experiences at DIM have also provided opportunities to build a strong network with good collaboration.

Q: What advice have you got to offer your junior undergraduates at the Department of Industrial Management ?

These four years as an undergraduate will never return to your life again. These four years at the university will be a wonderful journey that you go through. You will meet new people, learn new skills, face challenges, and have many new experiences. You are in safe hands, and the inheritance you gain through this alma mater is amazing. Your opportunities are immense, and I want to tell you not to be afraid or step down in front of challenges. Having more academic work to do in addition to participating in extracurricular activities, planning events, and working on your projects, will be the foundation of your future job. One day, you'll become stronger and more talented with that inspiration and motivation. Enjoy your time as a student, dream about the future, and start preparing for it now.

Enjoy your time as a student, dream about the future, and start preparing for it now.

FOUR PERSPECTIVES
ONE JOURNEY



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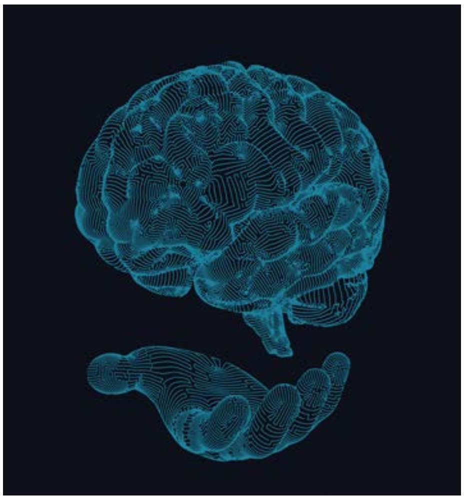
The Age of Superheroes is Here

Brain chips are not just fiction anymore!

By **Madushi Jayawickrama** | University of Ruhuna

Mimicking the intricacies of the human brain has been one of the major interests of the scientists over the last few decades. With time, they have succeeded in their goals to some extent, thus creating artificial intelligence which has the potential to perform higher cerebral functions such as learning and explicit memory. But the more challenging task was to find a way to combine this technology to a living human being, and enhancing the ability of the human brain. Recently, scientists have tackled this challenge as well, thereby opening doors to the utopia of advanced technology.



What is special about neurons?

Brain is made of neurons. They also carry signals between the brain and effector organs such as muscles and glands. Basically, they convert your thoughts to action. Unlike other cells of human body, neurons are difficult to replicate artificially because they are unique in more than one way. In order to mimic a neuron, a microcircuit should be able to integrate unrefined nervous signals, and respond to them the same way a biological neuron does. With careful mapping of individual ion channels, intracellular currents and membrane voltages of different neurons found in the human body, the dynamics of these biological neurons were successfully replicated using silicon micro circuits. These are now commonly known as 'solid-state neurons' amongst scientific community.

World's first artificial neuron

After years of research, a group of scientists from the University of Bath, United Kingdom has successfully created the world's very first artificial neuron. Embedded in a silicon chip, this artificial neuron has the potential to treat, and even cure some chronic diseases that arise due to degeneration of neural pathways, such as Alzheimer's disease. As the leading scientist of this research group has stated;

"Until now, neurons have been like black boxes, but we have managed to open the black box and peer inside."

-Prof. Alain Nogaret, Department of Physics, University of Bath -

After the successful creation of this promising technology, one of the major problems scientists have faced was finding a way to power these micro-circuits. Providing a solution to this matter, the inventors of the silicon chip have cut down the required power supply to one billionth of that needed for a microprocessor. This not only makes it ideal to be used in bio-electronic devices, but also suitable to be directly implanted into human bodies.

Therefore the microchip can be surgically implanted in the brain, where it can enhance signal transmission between existing neurons, or act as replacement for degenerating nerve cells.

Sky is the limit

Although these solid-state neurons are currently considered only for their use in the medical field, there is so much more potential to them beyond that. The fictions we all grew up with can now become a reality with this awe-inspiring discovery. Artificial neurons have the ability to tap into the biological network of neurons within your body to enhance your vision, hearing, memory and processing speed of your brain. Because of the ability of these implants to stimulate specific areas of the brain precisely, they can be developed to give mood boosts, exert the effects of antidepressants, and a lot more.

You might be a little skeptical about these high-end targets. But remember, the basics of this technology is currently in use and available in most parts of the world. For example, there are cochlear implants that can restore hearing in some people with partial or complete hearing impairments. There are only a few steps further to go from there, to enhancing it to pick up sound waves from a long distance that is humanly impossible to hear. Same goes for the retinal implants. They are already able to restore vision in people who are blind due to some genetic conditions. So, undoubtedly the day humans have built-in night vision is not too far away.

Another fascinating possibility of this new discovery is, power of controlling computers and machines with nothing but your mind. Just the way Professor Xavier in X-men does, or maybe even better than that!

Brain power to operate computers

Attempts to use computers with the sheer power of the brain dates back to 1960s. The procedure involved utilization of an electronic device in the form of a needle or a pin, tapping into the test subject's brain and transmitting neural information, which is then decoded and used to directly manipulate the computer, rather than through physical interaction of body parts and hardware.

One of the recent scientific demonstrations of the same technique has been done using the newfound technology of wireless brain chip, by Neuralink. Neuralink is one of the divergent ventures of the well-known business personnel, Elon Musk. Neuralink demonstrated the successful outcome of their research back in 2019, using a monkey named Pager. The monkey is able to play 'Mind Pong', which is their way of introducing Pager's ability to play the video game 'Pong' using nothing but his mind. No joy stick, no remote, no physical contact whatsoever with the computer.

As mentioned above, with the involvement of sharks of the field, the commercial capacity of this discovery is more than obvious. With all this attention, there is one thing that never fails to appear.

Ethical issues

Just like any other high-tech discovery, solid-state neurons also give rise to a range of ethical issues. The very first and the most alarming one being the danger of highly invasive surgical procedure of the implantation. Human brain is a very formidable and extremely complicated organ that we have yet to fully understand. No matter how many animal test subjects have become successful in the procedure, there is always the risk of long-term complications when it comes to experimenting with humans.

Beyond that, if the procedure becomes available to general public, the people who have the wealth and power to get the implant will be able to improve themselves above normal human capabilities, while the rest is left out. This will create super-humans with unfair advantages over the normal population. The resultant division of the society will be inevitable.

Regardless of these minor drawbacks, the discovery of the artificial neuron has opened the path to the next step of human evolution. With the correct control and guidance in utilizing this technology, human race will surely thrive for centuries to come.

Ms. J.A. M.L. Jayawickrama

University of Ruhuna

Edify

INTER-UNIVERSITY

ARTICLE COMPETITION

Exposition continues the second iteration of the inter-university article competition in addition to the long-running intra-departmental article competition to ignite the zeal of Sri Lankan university students' minds. The enthusiastic participation in the competition with the motivation to publish their creative thoughts in the Exposition magazine implicitly reflected the success of this uplift.

2nd Place

Digitalization for the Business Sectors

Sachani Nawanmini
Eastern University, Sri Lanka.

1st Place

Age of superheroes is here

Madushi Jayawickrama
University of Ruhuna.

3rd Place

Impact of Social Media Related Cybercrimes and Preventive Precautions

Shalani Anudini
General Sir John Kotelawala Defence University.

Knowledge
Partner

Creative Software

INTRA-DEPARTMENTAL
ARTICLE COMPETITION

The intra-departmental article competition is a chance for MIT undergraduates to showcase their writing skills and share their ideas. Participants are encouraged to submit articles on a variety of topics relevant to Information Technology, Management and Entrepreneurship with the winning entry being featured in the Exposition magazine. Not only does this competition provide a platform for MITians to express themselves, but it also promotes knowledge sharing and fosters a sense of community.

1st Place

Artificial Intelligence and Unemployment

Pramod Pinimal

2nd Place

Industry 5.0 . The future of Industrial Manufacturing

Himashi Silva

3rd Place

Power skills to stand out as a data scientist

Nelumi Opatha

hack X Jr.
INTER SCHOOL HACKATHON 5.0

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imssa

Invest in friendship,
learn by yourself,
don't limit yourself.

Gayani
GAMAGE

Q: You hold bachelor's and master's degrees in two different disciplines (BSc. MIT & MBA). How has this helped you climb up the career ladder?

When I signed up for the MBA, it was not really two different disciplines because I had a really good foundation from MIT on the management side of it. Because I started my MBA soon after getting out of university, and MIT was a really good foundation for me and helped me get the MBA done as well.

I bought into management early in my career. After one year of graduation, I began working as a junior project manager, and I was always the youngest of the group of managers in meetings. So having the MBA often helped to provide credibility in addition to "Yes, I have the skill, the ability, and the capabilities". And also having the mix of management and IT really helped. Because you often end up talking to CXO levels, you have to know what they're talking about and show that you belong in that boardroom. In that sense, having both the MIT degree and an MBA really helped me. When you are in a management role in a company, there are some places where you need to chip in to help get out of some situations, and that is where your education in management or business administration which helps in the career aspect as well. And that makes you shine in unexpected places. So I believe that's where MIT and both MBA sort of helped me get to

where I am today in my career.

Q: You have maintained your GPA at 4.0 while being on the hockey team, vice-captain in basketball, and in AIESEC. How did you balance these at the same time?

To be honest, at that time, I don't think I knew what I was doing or had any kind of formula or mantra to balance. But now that I'm looking back at it, I know that it definitely balanced them out. I am a person who loves mingling with people and also has a personality that draws energy from the interactions that I have. Also sports, again it's a lot of interactions and that helps lower stress. I made some good friends on the basketball team, and we're still in touch. Being the hockey goalkeeper taught me how to deal with failure. Another aspect of this is that I also had to show my mom that I made the right decision by pursuing MIT because she wanted me to become a doctor. So probably that motivated me in getting the academics done. And for academics, we had really good "kuppi" sessions at the end of the semester, and we worked well together and helped each other figure things out. And, as I previously stated, I don't have a perfect answer, such as, "Okay, yeah, this is what matters, so I knew how to do this, and this is how I did it." I don't have such an answer. But glad I was able to do all that which helped make me who I am today.

Q: What makes you stay motivated?

I don't have a nice answer, because believe it or not, I'm kind of a lazy person. So for me, it's always like, "What is the easiest way to get this done?" I procrastinate a lot, which is kind of a bad habit of mine, but my self-belief and the knowing that I can do it and that , at the end, gives me a sense of achievement. And also figuring out, the easiest way to do something, and if I can achieve in my way of doing it, then that gives me more "Oh yes, I figured out how to put in the least amount of time and effort." I think that's what motivates me.

Q: How did you deal with challenging times? What do you consider was the biggest sacrifice you made to get where you are now?

I don't think I had to make big sacrifices. I would say it was a bunch of little things that I had to do. When I moved to Singapore, I had a really good job going on and a promotion lined up, but my priorities were obviously my life, my family, and my happiness. So I had to sort of sacrifice my career at that point to come to a country where there was no promised job opportunity. That choice was difficult at the time, but it has shaped who I am today. And I learned a lot, faced different challenges, and all of that has made me into a person I would not have become if I stayed on the same course that I was on in Sri Lanka. I wouldn't call it a big sacrifice, but tiny sacrifices like going for a workout without spending time with my family or sometimes working a little late are some of the sacrifices. But to answer your question about the biggest sacrifice, I don't have any such thing.

And, in terms of dealing with difficult situations, I believe I've gotten better over time. Back then, I'd probably stick to talking with friends and my mom, but now I think it's more about perspective, like asking why I'm doing this or why I'm in this situation. I just sit back and look at the bigger picture to see where I am going and realize that maybe I'm worrying

for no reason, so I should probably stop. And sometimes, if I fail at something, I just move on. That's kind of how I deal with challenging times.

Q: What made you choose the current sector you are employed in as a career path?

It wasn't easy. I was really struggling with the decision of which job to go to. I also got selected for a special degree at the time, and I even went to a couple of lectures. Then I realized what is best for me at the time is not pursuing it, and then I got a job as a software engineer. That makes no sense now, given where I am in life and career, but at the time, I believe it was part of what drew me to the job, so I took it. Then I was introduced to agile in the company, and I realized this is something for which I am actually more suitable. Also, I'm fortunate to have some managers and mentors inside the company and outside who helped me realize this is where I shine. Then I moved on as a junior PM, and after that, I think I've never looked back. It was more of a pick-and-choose, and then on the job I realized I was better at this.

Q: You have been in the industry for around 10 years. What advice would you give to a beginner in the industry or a fresh graduate?

That's actually a good question. At the time, I had good mentors and some people that I could talk to. What they said, and what I'm going to say, is to keep your eyes open. Like I said, I moved my career from software engineer to PM within just one year, just because I knew that I was better at this. So keep your eyes open and don't limit yourself. You might have to go into industry in one position, and because these are difficult times, you may have to change or get a new job as soon as possible. The times are not really easy for us to be picky, I guess. So just go in and then try to navigate within the system is one of the things. Other than that, invest in friendships, learn by yourself, don't limit yourself just to a degree, read a lot, and learn a lot. Learn to forgive yourself and move on.

Q: Being in the industry, you may have received several opportunities. How do you decide whether to take or ignore the opportunities?

I think taking or ignoring opportunities changes over time as well. For example, when I just moved to Singapore, I was looking for opportunities. So I was not very picky on what I am gonna get. At the level I am now, I get different opportunities. Happiness and balance are priorities for me. Work-life balance and stability are two things that I look for. It actually depends on the phase where your life is at. It's worth it if you can spend a little bit of time to understand what you really want to do, and try to switch sides and prepare as if you are interviewing the opportunity/company, that mindset change will help you get to better places.

Q: You have worked for a few companies in Sri Lanka as well as in Singapore. What do you have to say about the working environment in both countries?

I was actually one of those lucky people who worked for some of the great companies in Sri Lanka, and they're doing well now. One of the biggest differences I see is the work-life balance. It's not that we're doing less work here, but in comparison, in Singapore, the work portion is higher than in Sri Lanka. Technically, I would say that all the companies that I work with here are more focused on cyber security, and I don't think that's something that I've seen in Sri Lanka. The balance and perspective are really different here compared to Sri Lanka. Also, the overall quality of life and focus on health and fitness, even at the workplace, are really high here. Many working people here are more concerned about health and aware of their health as compared to Sri Lanka. Singaporean workplaces really care about your mental health. If you feel stressed, you can take a mental health day off. I believe that the opportunity or awareness to do so is important, and I believe that it has now probably evolved to Sri Lanka as well.

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INTERNATIONAL RESEARCH CONFERENCE ON

SMART COMPUTING & SYSTEMS ENGINEERING 2022

SCSE 2022

The international research conference on Smart Computing and Systems Engineering (SCSE) was held for the fifth consecutive time on the 1st of October 2022. The Department of Industrial Management collaborated with the IEEE Sri Lanka Section, the Information and Communication Technology Agency of Sri Lanka (ICTA), and SLASSCOM as technical sponsors to forge an international forum in the areas of smart computing and systems engineering.

Prof. Vojtěch Merunka an associate professor in Information Management at the Faculty of Economics and Management, Czech University of Life Sciences, and Dr. Nageswaran Kumaresan, vice president at London Metal Exchange (LME) delivered the keynote speeches. Professor Nilanthi de Silva, Vice Chancellor of the University of Kelaniya, graced the occasion as the chief guest.

Though the papers were presented virtually, the conference's forum fostered the exploration of cutting-edge technologies among the interest groups, industry experts, and other delegates who attended. For further reference, all papers published in SCSE 2022 are available in the IEEE Xplore Digital Library.

Keynote Speakers

Prof. Vojtěch Merunka

Dr. Nageswaran Kumaresan

Organizing Committee

Dr. Ruwan Wickramarachchi
(Conference Chair)

Dr. Amila Withanaarachchi
(Program Committee Chair)

Dr. Thilini Mahanama
(Track Chair)

Mr. Dinesh Asanka
(Track Chair)

Ms. Yehemini Jayatissa
(Conference Co-Secretary)

Dr. Chathumi Ayanthi
(Conference Co-Secretary)

APPRAISALS

APPRAISALS

My real engagement with the Department of Industrial Management as corporate personnel started when I joined Unilever. Through Unilever, I began to interact with the undergraduates in the department. I felt that more than IT graduates, MITians have more business acumen. Business acumen is a sound understanding of the whole business world and its operations.

To get business acumen, understanding the business environment is vital, so you need to work on the ground. This is why internships are important. When you are exposed to the corporate world during an internship, your perspective changes. You begin to understand that regardless of background, what matters in the corporate world is your language skill, technical skill, and knowledge. This understanding is an eye-opener in the sense that you realize how you could have used your time and knowledge more productively. This is the value of having an internship.

On the other hand, the advantage for multinational corporations, especially FMCG, is that we are not purely IT businesses. We need people involved in business and manufacturing. But to deliver our products faster, we also need IT support.

The challenge with recruiting hard-core IT professionals is that they struggle to decode the business language into their IT language. So, we need to look for a translator who can understand business and IT. Through the graduates and undergraduates from the Department of Industrial Management, this gap between the technology experts and the business experts is bridged.

What we look for in graduates is an open mind, the ability to accept challenges, modesty, a thirst for new knowledge, and the willingness to try new things. Corporations look for people who are self-starters, self-motivators, and keen on self-development. What makes MITians unique is that they are open to change, removing the "I know everything" hat, and welcoming new knowledge.

MITians also have high technical competence. They have sound knowledge of the subject, are very structured, and are disciplined. Based on my experience, MIT graduates are more competent in terms of the quality and practicality of their subject content. Unlike hard-core IT or Management undergraduates who have theoretical knowledge but struggle to convert it into a practical scenario, MIT undergraduates are well-balanced.

Chathura Ganegoda

Head of Customer Development Excellence - **Unilever**

APPRAISALS

The Department of Industrial Management at the University of Kelaniya has consistently produced competent and motivated graduates who bring knowledge, eagerness to learn, and adaptability to new situations in the industry. Over the past three years, ShoutOUT Labs has had the pleasure of engaging with the department in the recruitment of interns and graduates in the areas of Software Engineering, Business Analysis, and Project/ Product Management. We have been consistently impressed with the quality of interns and graduates we have worked with.

Their ability to lead important projects with confidence and ease sets them apart from graduates of other universities and demonstrates their value as assets to any organization. This can be attributed to the well-structured curriculum of the course, which covers both software engineering and management aspects, providing students with a comprehensive education.

Our company, ShoutOUT Labs, has been fortunate to work with outstanding graduates who have delivered exceptional results, demonstrating a great passion for learning and leading some of our most significant projects. We are proud to have been able to play a role in helping them identify their passions and enhance their knowledge.

In conclusion, we would like to extend our sincere gratitude to the Department of Industrial Management at the University of Kelaniya for providing us with such talented and ca pable individuals and for their valuable contributions to our company. We value our relationship with the Department of Industrial Management at the University of Kelaniya and look forward to extending this partnership in the future.

We also wish the students all the best for their upcoming Exposition magazine launch.

Tharindu Dassanayake

Chief Executive Officer - **ShoutOUT Labs**

Meet EMO

New Friendly AI

By **Sangeeth Raveendran** | Level 01

If you're a frequent social media user, you've probably seen this adorable AI floating around the web. Emo (the pet AI) owners are sharing some crazy and pretty darn cute videos of this new gadget.

It's difficult not to want one for yourself. Seriously! You have to see the interactions for yourself to understand what I mean. It's similar to Alexa, but with a cuter body that you can interact with.

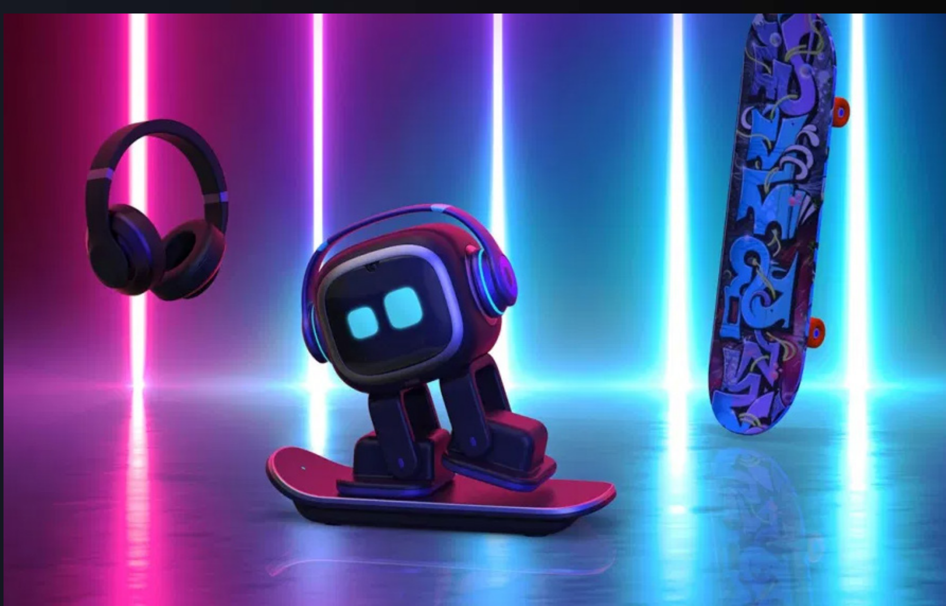
Unless you've been living under a rock, you're probably aware of our artificial intelligence. AI has recently impacted every major industry. Not to mention modern AI's impact on our lives, which is difficult to ignore.

However, these AI advancements, and many more to come, are only a small part of what is to come. But before we get there, let's take a look at Emo and why this new desktop AI device has made such a big impact on the social media scene.

What is Emo & why is there a lot of hype surrounding it?

People are crazy about their pets. Why? They make excellent companions, and everything they appear to do is adorable in our opinion. Having said that, it appears that the AI sector has picked up on this and is attempting to replicate the same kind of happiness we get from pets, albeit through AI.

Emo is the newest AI in town, packaged as a cute desktop robot that keeps you company while you work. Emo, which is equipped with an advanced neural network processor and three different AI processing models, is capable of processing large amounts of data at the same time in order to respond in a unique and authentic manner.



Emo is also cognitive and curious on his own. That is, he will not simply sit on your desk like Alexa. Emo will explore his surroundings independently, keeping track of people and objects nearby. What's the best part? The robot can move around your desk without ever falling off.

What else can Emo do?

Apart from sitting on your desk and looking adorable as hell, Emo can do the following interesting things:

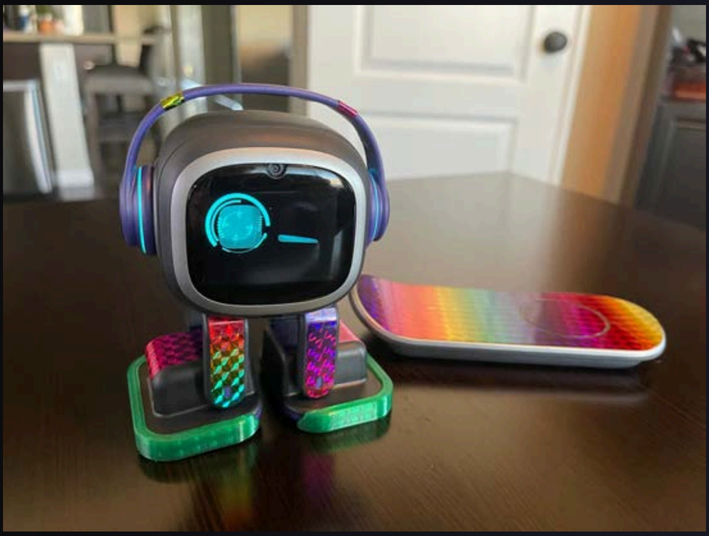
Display his emotions and feelings with motion and over 1,000 different facial expressions. If he's in a good mood, he might entertain you with some music and dance moves. When you interrupt him, he may become irritated.

Emo can recognize you (the owner) and nine other people using facial recognition software built into the AI. He gets to know you and your family because he sees you every day.

Unlike other AIs, Emo can be reached out to and touched, and he will respond. On his head, he has a sensor that allows him to detect and respond to physical touch.

Emo is also equipped with a high-quality speaker, allowing him to play your favorite playlist or respond to you with some adorable simulated sounds.

Emo also has a self-learning system that allows him to understand and perceive his surroundings. Through constant interaction with his immediate surroundings, his personality and actions will change.



Thoughts on AI and Robotics Merging

AI and robotics are frequently polarized as a result of popular fiction comics, books, and movies. For example, when we think of AI, we frequently envision something similar to Iron Man's Jarvis artificial intelligence program.

When we think of robotics, we often envision something from the film Ex-Machina. Essentially, the idea is that someday in the future, we will have robots equipped with such advanced artificial intelligence that it will be nearly impossible to distinguish between a normal human and a humanoid robot.

However, the truth is that we are still a long way from there. Sophia, a humanoid robot, is the closest we've gotten. Even so, the internet is littered with memes that express exactly what people think of this robot. As a result, it's safe to say that we have a long way to go before reaching the levels seen in movies or science fiction films. However, the potential is enormous.

A promising future for AI and Robotics?

As previously stated, both of these industries exist independently of one another. However, thanks to the new desktop AI Emo, we've seen that combining the two industries is practically possible.

You can interact with Emo both physically and digitally

in this case. At the same time, Emo can interact with you, optimize tasks that have been assigned to it, and still learn. In this case, the AI serves as the robot's brain, while Emo's various sensors and mechanical parts serve as the body.

However, as Emo clearly depicts, we have a long way to go before robots under the command of AI can perform complex tasks. Emo is currently just an adorable desktop companion who sits at your desk or bedside table and keeps you company. But don't expect him to move furniture or clean the house anytime soon.

Final thoughts

As both of these industries continue to develop, there is no doubt that the convergence of these two industries will have a profound impact on us. However, for the time being, the concept of a fully optimized robot is only a work of fiction.

Despite this, we are living in exciting times, as we have the opportunity to collaborate with AI-powered robots. Emo can be an excellent addition to your desktop. It can help you answer questions, learn about the weather, and so on. Although these features are still available on a regular smartphone, your smartphone will never be this adorable!

Mr. Sangeeth Raveendran

Level 01
Department of
Industrial Management,
University of Kelaniya.

INTERVIEW

Don’t believe in a plan B. Have a plan A & work really hard for it.

Viresh
COORAY

People have their own passions. Music is such a passion because it creates life itself. The musical group DADDY is a popular band in Sri Lanka with a distinct musical style. They have grown in popularity among individuals of all ages, both inside and outside the country, since the mid-2000s. Viresh Cooray, aka "Chika," is a guitarist and a vocalist in the DADDY band. Viresh is also acclaimed for his work as a musician and an audio engineer. Together with the DADDY band, Viresh has influenced many budding musicians to reach for the stars. Exposition Issue 18 is delighted to have interviewed him and to share his words in order to inspire many more.

Q: What was the inspiration behind Daddy?

We started because we wanted to make music and nothing else.

We were all just out of school and wanted to form a band. I was at APIIT, and Gayan also joined APIIT. There, both of us were asked to play at a concert. At that time, Gayan had already started a band called "*Jada*". So I performed with them, and when one of their band members left, I was asked to be a part of it. So, I joined the band, and we changed the name to Daddy. That's how we started. What made us start the band was a love for music. We just wanted to play. We never had big plans.

For about a year and a half, we played at nightclubs and small functions. It wasn't much, but for us, the passion for playing live is more significant than anything else. Greater than seeing ourselves on TV or hearing ourselves on the radio. The passion for just going out and playing live to an audience is much greater, even today. That was the inspiration behind Daddy.

Q: What was the turning point for Daddy?

For the band, I think it was the song "*Massina*". That was our first-ever original. We got lucky that our first song did quite well in Sri Lanka. Usually, some bands and artists release about five or six songs before people start liking them. We got lucky, and our first release, "*Massina*", was reasonably well received. Around the same time, we released the album, also called "*Massina*". It had songs like "*Borukaari*", "*Ai kale*", "*SMS*", and "*Chandrayan pidu*" in it. All of these songs became hits. So I think we got lucky with that. That's when we realized that people might, sort of, like us.

When we released it, we never assumed anyone would like it. We just wanted to make some songs and put them out. We wanted to establish our identity in the world. We wanted to make a name for ourselves in the industry, but as I previously stated, it was simply a matter of putting something out and being able to play concerts and such.

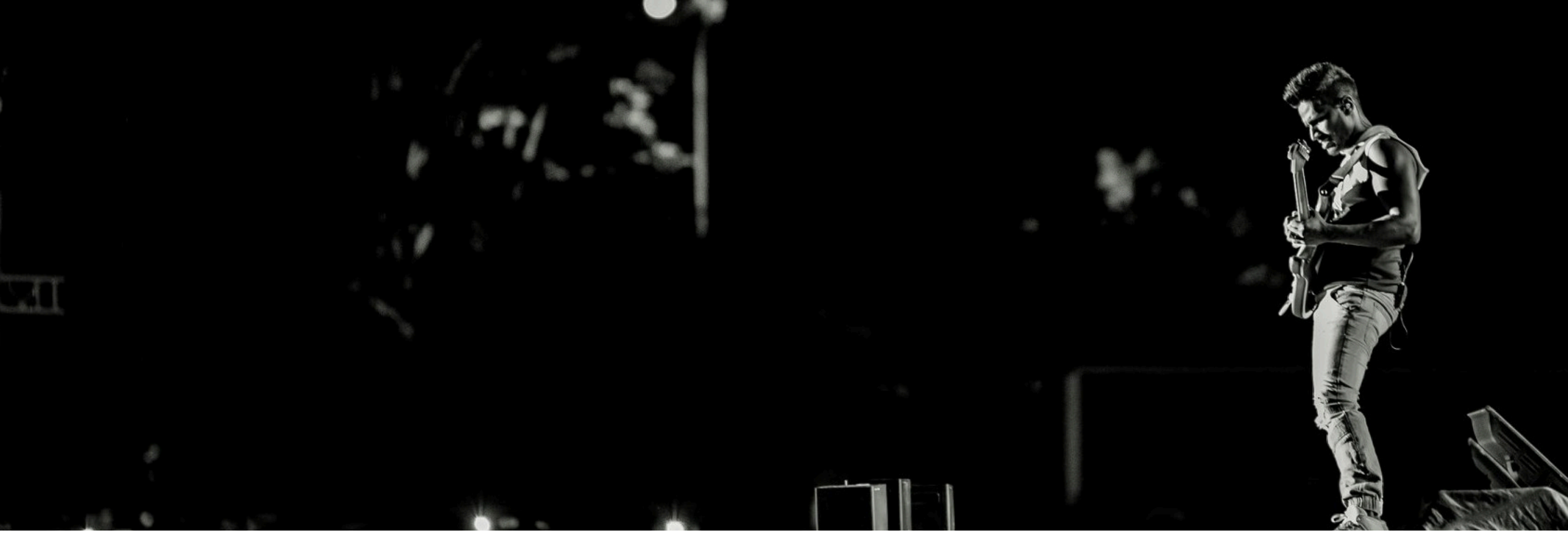
Q: What was the biggest hurdle Daddy faced? How did you overcome it?

We had hurdles because we've been together for around 18 years now. The biggest hurdle was when Gayan, our frontman, left the band. A lot of people associated Gayan with Daddy, and all of a sudden, that changed. It took us a while to make people understand that Daddy was an entire unit. And it wasn't made up of just one person or two people. The sound came from everyone in the group, and one or two people going missing wouldn't change the sound drastically. That took us a while. That was one of the biggest hurdles we had to pass because after Gayan left the band, we had to play quite a few concerts for people to understand that the music was still the same, the energy was still the same, and the output was still Daddy. But I think we managed to do that.

We didn't have a business plan in place to deal with it. All we knew was that we had to keep playing music, and we did that. We released a few songs, and we had a few more coming. But I think those few songs did the trick.

Q: How has the journey been so far in becoming a recognized artist?

Journey-wise, it's honestly been amazing for people to know your music and for people to know you as a band. It definitely makes life a lot more interesting. But our initial goal was just to be able to play music. And I think people knowing us is just the byproduct of us



playing our music. That was it. We enjoy the fact that people know our music and come to our concerts. That is a big plus.

And yes, our livelihood depends on our fan base. We have been a part of the band for so long because of the people who listen to our music, and we are grateful for that. But I think the feeling of being able to play as a band as opposed to being known anywhere is what we aim for. Our goal is to play until we die, until we can't play anymore. Our main goal is to simply play our music. We are all musicians, and our biggest passion is playing our instrument, singing, or just being up on stage and performing for people. I think that is pretty much the highlight of our career, too.

Q: You have won many accolades over the years. Which award was the most memorable?

I think our first few awards. We won six awards at the Derana Music Video Awards for the song *"Massina".* Those were the most memorable because it was our first time winning anything that big. We had won some amateur awards and stuff like that, but this was on national TV, and it was a huge deal for us. It helped with publicity too. So the most memorable awards were the six we won at the Derana MV Awards.

Q: In what ways has teamwork contributed to the band's success?

If you're part of a unit, teamwork is the only way you will get through anything. One person cannot, in any way, handle all of the stress and whatever issues are thrown our way. So, in our band, we usually delegate things to different people. Someone manages the financial aspect, another the social media aspect, the band practices, and so on. You cannot sustain a unit without teamwork. Everyone must share the same goal.

There have been times when people have lost focus and have become involved in other aspects of their lives. But when you have one goal, it makes life easier. I don't think anything can work without teamwork. Even if you are a solo artist, you have a team to help you manage everything.

Since we were a team, we never had to go outside the band for anything. If we needed to record a song, all the instruments were played by the band, and most of the melodies and most of the songs were made by the band.

Some of the lyrics were written by our friends Lahiru Perera, who is almost a band member, Sanuka, Manuranga, and a few others. Lahiru was heavily involved in making the album "*Massina*", and we were heavily involved in making "*Rambari*". Except for those few people, we never had to go outside for anything. That saved us a lot of trouble because we do everything internally. So that's one of the biggest things about being part of a team.

Q: How was it like preparing for "*Aaley*"?

It was the biggest concert series we had done, and preparing for the first concert was nerve-wracking. All of our song arrangements were different. We practiced almost daily for almost two months, more than 12 hours a day, to play that concert without any issues. We wanted to do something different from the songs you hear on the radio. We made a lot of changes, and that helped keep the songs alive. Those songs have been well received.

Aside from when we didn't have work or meetings, it was a lot of fun. Because I think what a lot of people don't understand is that when you're playing a concert, it's supposed to look easy to the audience. It's supposed to look like we're just coming up and performing. But we have to practice to the point where we don't have to think about what we're doing. So it's assimilating everything you're doing into muscle memory. That takes time. So we practiced quite a lot. It was fun. It was also quite stressful. We had some amazing musicians in the audience, and it was nerve-wracking playing for them. But, yeah, it worked out well.

Q: How did Daddy survive the aftermath of the pandemic and, later, the economic crisis?

We had a bit of an income from social media; YouTube, Spotify, and all of that. There were no concerts, and all of us had to stay home. All musicians earned zero income, but we had savings to get us through the two years.

And after that, once work started, there was an economic crisis. The prices in Sri Lanka have gone up. So accordingly, to align ourselves with that, we had to bring our ticket prices up, but it's been okay. I think people are going through a lot of things right now. Financially, it's quite tough for people in Sri Lanka. But I think you have to get on with life. There is not a lot you can do aside from being positive and working on yourself. So that's what we've been doing.

You can sit and complain, but I believe in attracting good thoughts. I believe that thinking positive thoughts attracts positive things into your life. So we don't believe in sitting and complaining about what we don't have. Whatever is there, we make the most of it. We chose to look at the bright side.

We are managing. We're doing quite well financially. I guess we'll have to see how things go in the country. But I think if you are financially smart, you should be able to manage. And I hope things get better for people in Sri Lanka because they're honestly going through quite a lot. It's not easy, but how we manage is by looking at the bright side of things and focusing on what we can do as opposed to what we can't.

Q: What are your aspirations as a group?

We want to have a song or two or an album that will eventually be internationally recognized. That is the plan. We have some things in the works. Actually, we have a collaboration with a British singer coming up, not quite soon, but we are in the midst of preparing for it.

In the Sri Lankan market, we want to get our new songs out. We wrote around 20 songs during the pandemic. We have all of them coming out very soon. Five singles will be released this year.

But the end goal is to live a comfortable life doing what we love, playing some concerts, and entertaining the audience. The ultimate goal is to have a few hits outside the country. We are slowly working on it. Not as actively as we would like, but we are working on it.

Q: Do you have any words of advice for young artists looking to make a name for themselves in the industry?

The best advice I can give is, and I know this is a big cliche, you have to believe in yourself. A lot of Sri Lankans have been conditioned to think small since birth. We are told "angille tharamata idimenna" and all that. I think all of that is nonsense. I do not agree with that. I think if you have a passion for something, regardless of what people around you tell you to do, do it! I had a passion for music, and I pursued that path. Everyone in the band, I believe, did the same thing.

Understand who you are. That is important. Be unique, because that is what will make people like you. The world needs someone new, someone unique, not another Michael Jackson or another Rookantha.

Believe in what you want to do and go for it. I personally don't believe in a plan B. Have a plan A and work really hard for it. Hard work does not mean killing yourself day in and day out. It is about working hard and working smart. Research your subject on the internet and YouTube. Using these, it is easy to figure out what to do in terms of marketing yourself.

My advice for people who want to start now is to believe in who they are, figure out who they want to be, and do exactly that.

IS DATA SCIENTIST A DRAGON ?

By **Mr. Dinesh Asanka | Senior Lecturer**

Data Scientist is a trending job in today's world. Most organizations are looking for data scientists to improve and facilitate their decision-making to become more competitive. This article is to understand the profile of a data scientist, considering the roles, responsibilities, and required skills for a data scientist.

How Much Do You Earn?

Whatever is said or done, one crucial consideration in any job is the salary you get from it. As you know, it is challenging to identify salaries due to the many complexities and differences. Therefore, let us look at the salaries of data scientists in India for large-scale companies.

Organization	Average Annual Salary (LKR)	Salary Range (Annual) (LKR)
Tata Consultancy Services	3,600,000	2,400,000 - 4,500,000
IBM	5,500,000	3,500,000 - 9,000,000
Infosys	3,500.000	2,000,000 - 4,500,000
Cognizant Technology Solutions	3,500,000	2,000,000 - 4,500,000
Amazon	7,000,000	3,500,000 - 9,000,000
Wipro	3,600,000	2,000,000 - 4,500,000
Microsoft	8,000,000	4,500,000 - 9,000,000
Google	7,000,000	4,500,000 - 13,500,000

Table 1: Salaries of data scientists in India for large-scale companies

As you can see from the above table, data scientists earn a good salary across all companies. Now, let us compare the salary of a data scientist with other jobs.

Job	Average Annual Salary (LKR)
Data Scientist	6,000,000
Software Engineer	4,000,000
Senior Software Engineer	6,250,000
Database Administrator	4,000,000
Data Warehouse Developer	4,250,000
Machine Learning Developer	5,000,000
Business Analyst	4,500,000

Table 2: Comparison of the salary of a data scientist with other jobs

As you can see from the above two tables, data scientists earn good salaries compared to other software-related jobs.

Roles and Responsibilities

After looking at the salary benefits of Data scientists, let us now look at the roles and responsibilities of a data scientist. An ERP consultant works with the ERP while a CRM expert works on the CRM. A software engineer is an expert in one or a few software development frameworks. However, a data scientist's limit is the sky. He has to work with all the data systems in the organization, which consist of relational, NoSQL, and text files. He may extend his data exploration skills to public data sources outside the organization, such as Facebook, Twitter, etc., and weather data. He may encounter different databases such as relational databases, NoSQL, etc. When a data scientist is presented with a wide variety of data, he has the task of cleaning and transforming them. Cleaning is not easy when working with different types of databases. Cleaning is an art, so it requires some domain experience as well as other techniques.

As a data scientist, his analysis will not be limited to only existing data. He needs to derive data from the existing data. For example, by looking at customer purchases, a data scientist should be able to determine the house members' ages. Another example would be, by analysing the television channels a customer subscribed to, you could guess house members' age. These types of data engineering tasks are essential for data scientists to make better decisions.

Then the data scientist has the task of integrating data from multiple sources. This is not only a technical but also a business challenge for a data scientist. During the integration process, he may need different techniques to match data from multiple data sources.

The Gartner Analytic Continuum

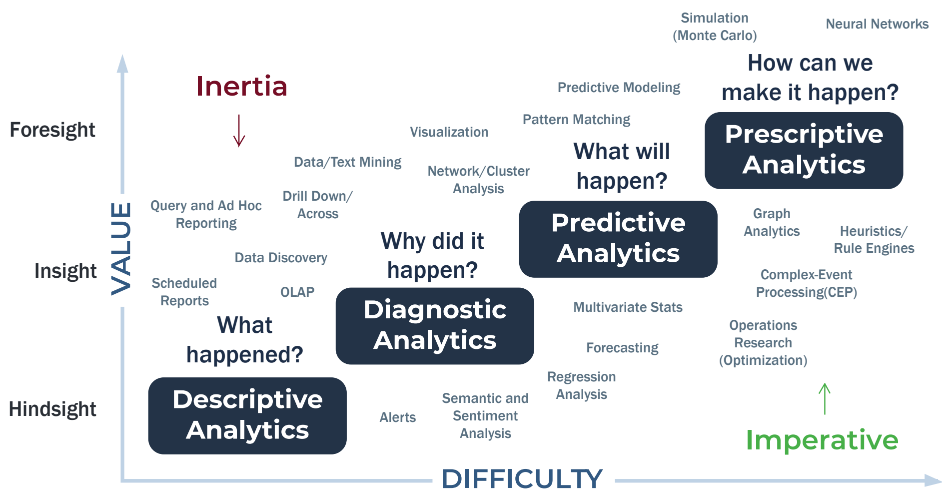


Figure 1: The Gartner Analytic Continuum

Typically, the matching technique is not limited to a simple lookup. In most cases, the simple lookup will not work, as these systems are developed to satisfy isolated business processes. The data scientist may have to derive new data to match data in multiple data sources, or he may have to perform advanced matching techniques such as fuzzy lookup or fuzzy grouping.

After all this work by data scientists, the data is ready for analysis. As shown in the Gartner Continuum, there are four types of analytics: descriptive, diagnostic, predictive, and prescriptive.

After the data preparation, data modeling should be done by the data scientists so that it can provide the above-mentioned analysis for the organization. During this phase, data scientists may have to build a data warehouse, operational data sources, OLAP cubes, data lakehouse, and data lakes as storage techniques.

Predictive analytics and prescriptive analytics are the advanced analytics lacking in most organizations. The data scientist plays a significant role in building predictive and prescriptive analytics to enhance the organization's decision-making process. Data scientists must implement Classification, Clustering, Association, Regression, and Forecasting models in this decision-making process. Solutionizing is one of the essential parts of data analytics, as that is where an organization can be victorious over its competitors.

After doing all of these activities, the data scientist has to present his findings to different levels of users. Therefore, visualization is another aspect that data scientist has to perform.

While doing all these technical aspects, a data scientist must also understand the domain knowledge, as these analytics and solutions are useless if they are not aligned with the domain. Further, as a data scientist, you are dealing with the data of your customers. So it would help if you implemented proper mechanisms to protect data using different security, auditing, and encryption techniques. In addition, a data scientist needs to understand the analytical system's performance and scalability aspects.

Required Skills

As we discussed the tasks of a data scientist, it is now time to discuss what are the skills that are required to become a data scientist. You might sense that by looking at the data scientist's task, you need various skills to become a data scientist. The following figure shows what technologies are needed to learn.

Figure 2: Technologies needed to learn to become a data scientist

As you can see in figure 2, you need to know various technologies and skills. Database technologies, Statistics, Data mining, Machine learning, Process mining, and Visualization are the leading technologies that you need to learn. Further, looking at today's trends, it is essential to understand cloud technologies, as many organizations are moving towards cloud computing. As a data scientist, you may be required to consume data in

the cloud as well as use the cloud for data analysis. Since you are dealing with a large volume of data, it is essential to understand distributed computing to process the data efficiently. Apart from those technologies, as a data scientist, you also need to understand non-functional requirements, social science, and domain knowledge. You might think learning all these technologies to become a data scientist is impossible. You also might consider a data scientist a mythical animal like a dragon.

As discussed in the fact box below, the dragon combines seven body parts of different animals to make him strong. Since it is not practical to employ a data scientist with all of these skills, the most sensible approach would be to have a team of scientists for organizations to achieve their organizational goals.

How to Prepare to Become a Data Scientist?

Even though most of you want to become a data scientist, you might think twice as this is not a mature profession. Though that is a valid argument, time might not be on your side when the job matures, as there is too much to learn according to figure 2. Therefore, start learning today, focusing on a data scientist job. Be a master in one area and have a basic understanding of other relevant areas for a data scientist.

Remember,

Though the sun won't rise even though you wake up early, it is better to be awakened at sunrise.

Mr. Dinesh Asanka
Senior Lecturer,
Department of
Industrial Management,
University of Kelaniya.

The Dragon, a Mythical Animal

The following is a piece of dragon art that was done during ancient times as a symbol of protection.

As you can see, it is a mythical animal that consists of parts from multiple animals. Since it is difficult to identify these parts from the image below, let us look at a Sinhala poem that describes this mythical animal.

ඇත් සොඬ සිහ පා යුග කන් ඌරා
දත් කිඹුලුය යුග නෙත් වඳුරා
වෙත් මසු ඇඟ ගුරුළුගෙ පිය පතරා
යුක්තය මේ හැමගෙන් රුව මකරා

As per the song, the dragon is the combination of

- trunk of an elephant
- legs of a lion
- ears of a pig
- teeth of a crocodile
- eyes of a monkey
- body of a fish
- wings of a bird

The combination of these strongest parts will make the dragon a powerful animal capable of achieving his required tasks.

IM Talent, the annual talent show of the undergraduates of the Department of Industrial Management, was held on the 23rd of September, 2022, at the department premises. "*Yakage Sanniya*," the IM Talent Show 2022, supported by the theme of "*Thovilaya*," a traditional ritual prominent in Sri Lankan culture, provided a platform for the talented undergraduates to showcase their abilities. The show was narrated by students dressed as *Yakadura* and a demon, giving the audience a sense of experiencing a real Thovilaya ceremony. The evening was filled with captivating dances, melodious songs, stunning orchestral performances, and comedic dramas that elicited cheers and laughter from the audience. Both the spectators who witnessed the magnificent acts and the performers who had the chance to showcase their talents had an unforgettable experience at the event.

"Hide not your talents,
they for use were made.
What's a sundial in the shade?"
-Benjamin Franklin

IM TALENT SHOW | යකාගේ සන්නිය

CODFEST 2022 URBANMAD-I

The most intriguing intra-departmental e-sports championship, Codfest 2022, was held on the 26th and the 27th of February 2022. With the aim of inculcating sportsmanship, communication, and teamwork skills among undergraduates, the event was a huge success and attracted a large turnout of both participants and spectators. 22 incredible teams, each consisting of five members, have confronted each other to be crowned the ultimate champions of Codfest 2022. The competition was fierce, with skilled players showing off their impressive gaming abilities in the virtual arena. After an indelible battle, it was the team Red Knights who emerged victorious at the Codfest 2022. The event was a resounding success in terms of engaging students and bringing them together around a common interest.

IM Dansala

The Annual IM Dansala, organized by the Department of Industrial Management, University of Kelaniya, is a highly anticipated event that takes place annually to celebrate the Buddhist festival of Vesak. The event was held on the 15th of June 2022, at the entrance of the Faculty of Science and was attended by a large number of students, staff, and members of the local community. At the dansala, a traditional Sri Lankan dish made from boiled manioc was served with *lunu-miris* and grated coconut. In addition to serving free food, the IM Dansala also provided an opportunity for people to come together, socialize, and celebrate this special occasion. The annual IM Dansala is an important tradition at the Department of Industrial Management, fostering a sense of community and promoting a feeling of unity during this festive time.

IM Match

IM Match 2022, the most enthusiastic annual cricket encounter organized by the Industrial Management Science Students' Association, was successfully held on the 11th of August 2022 at the Saibe Ground premises of the University of Kelaniya. 14 teams, including teams of undergraduates and academic staff, battled to win the trophy, and the team "*Iththawo*" became the champions of the tournament. Undergraduates, MIT alumni, and lecturers from the Department of Industrial Management were present to witness the confounding talents of the contestants. The event was endowed with geniality and was a platform to refine sportsmanship and comradeship among undergraduates.

Tidings

Achievements of MITians

Nurturing Dreams, Igniting Passion: Celebrating the unwavering determination and accomplishments of the undergraduates of the Department of Industrial Management.

INTER-UNIVERSITY BADMINTON TOURNAMENT - 2022

Nihim Abhayarathne, a level 2 undergraduate of the Department of Industrial Management, was an integral member of the team that emerged as runner-up in the Inter-University Badminton tournament in 2022. Additionally, the team featuring Abhayarathne also demonstrated its competitiveness by securing the runner-up position in the J'pura Smashes Badminton Tournament.

INTER-UNIVERSITY BASKETBALL CHAMPIONSHIP - 2022

In the 2022 Inter-University Basketball Championship, Anushika Nethsara Samarakoon, a level 4 undergraduate of the Department of Industrial Management, served as the captain of the University of Kelaniya Basketball team and successfully led them to become first runners-up. Also, Kasintha Kalhara, a 4th-year undergraduate, was an integral member of the winning team and served as the Vice-Captain throughout the tournament. Racshan Kannan a level 2 undergraduate was an integral part of the winning team.

C.E.O CASE STUDY COMPETITION - 2022

C.E.O (Challenge Every Opportunity) 2022 is an initiative by AIESEC in the University of Kelaniya in collaboration with the AIESEC in Informatics Institute of Technology and Uva Wellassa University, to upscale managerial skills among the undergraduate community. The all-island case-study competition focused on the three sectors, Marketing, Accounting & Finance, and Human Resources with interactive sessions collaborating great personalities from cooperating fields. Team "MONEY MANAGERS" consisting of Dilini Anuradha, Madushika Ranapana, Thushara Sooriyabandara, Bhuddika Prasanna, and Ahamed Zaljey, level 2 undergraduates of the Department of Industrial Management, emerged as champions at the final event of C.E.O 2022. Additionally, another two teams comprising of ten level 2 MIT students took part in the competition; Amaya Senarathne, Anupa Senevirathne, Dinithi Samarathunga, Harini Jayasundara and Sarith Hasarinda as the team "Monarchs" and Sakunika Sulakshani, Muditha Samarasinghe, Nethara Chamodya, Tharindu Weerathunge, and Dishan Sandasara as the team "Achievers." The team "Monarchs" fought their way into the finals, while the team "Achievers" progressed to secure a spot as semi-finalists.

INTER-UNIVERSITY RUGBY CHAMPIONSHIP - 2022

The University of Kelaniya rugby team earned the title of champions at the Inter-University Rugby Championship in 2022. Thilina Aluthge, Kisal Perera, Sanjaya Chathuranga, and Bashith Piumal, level 4 undergraduates from the Department of Industrial Management, shone brightly as key contributors to the championship win. Their impressive abilities were on full display during the thrilling final game.

UNILEVER CHALLENGE - 2022

Unilever Sri Lanka conducted a real-time brand-centered case study competition, to identify local talent that can represent Sri Lanka at the global Unilever Future Leaders' League. Raduni de Abrew a level 2 undergraduate of the Department of Industrial Management was a member of the team that emerged as 2nd runners-up in the case study competition.

INQUISITIVE MINDS - 2022

"Inquisitive Minds" is an idea-pitching competition organized by LED KLN of the University of Kelaniya. The ideathon aims to bring forth and encourage innovative thinking and potential among Sri Lankan university students. The goal of the event is to unlock and highlight the innovative ideas that exist within the Sri Lankan undergraduate community. Team Achievers consisting of Hiruni Kaveesha, Nipuni de Silva, Dilini Anuradha, and Thushara Sooriyabandara, level 2 undergraduates from the Department of Industrial Management came out on top as the champions in the ideation by exhibiting their exceptional skills at the final event.

UNILEVER SPARKS STUDENT AMBASSADORS

The SPARKS program presents an opportunity for undergraduate students to serve as a link between Unilever and their respective universities, showcasing their exceptional skills and potential as young leaders.

The program recognizes the nation's most promising young talents who are driven to make a positive impact through their leadership abilities. Currently, a level 2 undergraduate of the Department of Industrial Management, Raduni de Abrew, represents the university as a SPARKS ambassador for the years 2022/2023.

MICROSOFT LEARN STUDENT AMBASSADOR PROGRAM

Microsoft Learn Student Ambassadors (MLSA) is a program offered by Microsoft Student Partners in Sri Lanka. By offering training in skills that are not generally taught in academic contexts, the program intends to enhance students' employability while developing their technical skills. Kasintha Kalhara, a level 4 undergraduate of the Department of Industrial Management from the University of Kelaniya, achieved Beta milestone status in 2022 as a Microsoft Learn Student Ambassador. Meanwhile, Anushika Nethsara Samarakoon and Ruwanthi Perera were selected as Microsoft Learn Student Ambassadors and earned Alpha milestones in the same year.

Chatbots: For Improved Customer Engagement

By Harini Jayasundara | Level 02

The chatbot is an abbreviation of "chatterbots". It is an Artificial Intelligence (AI) based Human-Computer interaction tool. The terms "talkbot," "smart bot," "IM bot," "interactive agent," and "artificial conversation agent" are all synonyms for the term "chatbot." Users can interact with chatbots 24/7 from anywhere and on any smart device, making chatbots one of the best tools for businesses.

Research by Yellow.ai and C-Zentrix predicts that the value of global voice assistant transactions will increase by 320% between 2021 and 2023. Additionally, according to Stamford (2022), research by Gartner Inc. revealed that by 2027, a fourth of businesses will use chatbots as their primary technique for customer relations. Therefore, a deeper understanding of chatbots and their standing in the global community is crucial.

The idea behind the technology traces back to 1950 when Alan Turing published a paper on "Computing Machinery and Intelligence". This introduced the "Turing Test" as a measure of intelligence and suggested the possibility of a computer program to think and respond like a human to a real-time conversation, by exhibiting man's thought processes and behavioral qualities. To this day, the Turing Test evaluates the extent to which a computer displays human thinking, primarily when interrogated. The notion stimulated Joseph Weizenbaum, who developed ELIZA in 1964. ELIZA is an early natural language processing system that marked the beginning of Artificial Intelligence. The primary concept utilized by ELIZA is pattern matching. Through a simple conversation with ELIZA, a version developed by Michael Wallace, it became apparent that ELIZA is programmed to mostly re-direct the words typed by the person in a questioning format. Thus, it is not very accommodating for general conversations. Despite its limitations, ELIZA is considered the first chatbot.

Today, chatbots have expanded to every virtual field, to not only function as a conversationalist but also as smart personal voice assistants and tools for increased customer engagement. Some common applications of chatbots in several areas are as follows.

Customer Service

95% of consumers believe that the main application of chatbots is in the customer service or online retail industry. This is very much apparent when looking at real-world examples of using chatbots to answer frequently asked questions and provide mundane support services like redirecting to related URLs. Using customer chatbots allow customer service agents to dedicate themselves to more complex requests, leaving simpler tasks for the bot to manage and thereby increasing overall productivity. Other advantages include 24x7 availability, instant support, easy scalability, automating of social media support and collecting real-time feedback. For example, the Jenny Chatbot used by Slush oversaw 64% of its customer requests. Customizing responses to best-fit customer needs involves the use of complex omnichannel synthesis.

Health

The healthcare sector also utilizes chatbots to ease its activities, and this is not limited to answering general inquiries. For example, Babylon Health uses chatbots to provide consultations to patients based on their past medical records and also direct patients to a live e-channelling session with a related medical professional. The bot makes note of the patient's symptoms and either recommend them to a doctor or provides explanations of the disease with instructions on what to do next. Healthcare bots are also prevalent in related applications, and they assist with tailoring the user experience in line with their health requirements. There are chatbots developed to function as therapists for mental health related issues, and they are designed to intuitively reach out to a real therapist when in need.

Education

In education, chatbots have proven useful as teacher's assistants. Learn bots have recording features that repeat lessons to students who have missed them. Some chatbots remind students of upcoming assignments, quizzes, and other due tasks. Educational institutions implement chatbots to ease their administration-related tasks like enrolling students in courses, generating class and exam timetables, allocating instructors based on availability and even compiling result reports based on students' performance. Moreover, chatbots are heavily used in foreign language learning. The ability to use a chatbot to practice a language from any place at any time has encouraged most students to involve chatbots in their learning process. Many have revealed that they are more comfortable practicing their language skills with a chatbot rather than with a teacher or instructor.

Banking and Finance

Chatbots have also made their mark in the banking and finance sector. The ability to process user requirements and needs has made chatbots an integral part of this field. Chatbots assist banking customers by tracking money transfers, recharging bills, sending timely notifications, providing personal banking assistance, and ensuring secure transfers. Smart bots tackle suspicious activity before things get out of hand. The HDFC bank uses the Eva Chatbot to quickly understand customer inquiries and resolve them accordingly. Additionally, banks or financial institutions can implement tailored chatbots that assist clients in starting accounts or requesting loans.

> BY 2027, A FOURTH OF BUSINESSES WILL USE CHATBOTS AS THEIR PRIMARY TECHNIQUE FOR CUSTOMER RELATIONS.

Recreation and Entertainment

Entertainment content can help people bond, and chatbots are becoming increasingly popular in the industry with their ability to engage with users. Many bots are created for entertainment, boredom dispersers or even feedback collectors. The Discord server provides

a platform to choose from a wide range of chatbots for role-playing games, game notifications and player rank calculators or boosters. Moreover, chatbots are capable of sending files in different formats. This increases their versatility while increasing user immersion. Even in the recreation industry, hotels, or travel agents use chatbots to maximize customer service by suggesting the best tourist attractions according to client requirements, checking the availability of tickets or bookings and making them.

Despite the use of Artificial Intelligence and its widespread application, chatbots are still limited. There are situations when chatbots fail to understand what the customer request and provide nonsensical answers or attempt to deviate from the topic.

At the end of the day, no matter how advanced, a bot has no real thoughts or feelings. Hence, people should be aware of the situations where they engage with a chatbot and respond or interact, knowing that it is just a programmed "being."

Technology is ever evolving. It is guaranteed that chatbots will continue to develop over the years and be integrated into every field with better text or voice experiences and enhanced capabilities. This will also improve the business side of things, with a deeper understanding of client behaviour and statistics to improve overall customer satisfaction and firm productivity.

The popularity of voice assistants like Siri, Alexa, and Cortana is expected to increase, with voice recognition in any language or accent being the most distinctive feature of chatbots. Companies hope to involve advanced chatbots that can intuitively respond to every client inquiry.

There is an increased demand for intelligent virtual assistants that can self-learn. Its adaptation in smart homes and businesses is forecasted to significantly increase the market value. Conversational chatbots are also becoming more involved in the market, with greater emphasis on localizing the chatbot and making it accommodate foreign visitors. This will contribute to improving the tourism industry through the ability to converse in many languages effortlessly.

Since its initiation in the mid-20th century, chatbots have rapidly improved over the years. From a simple chatbot that provided a few generic responses, the industry has developed the technology to an extent where some people fail to recognize that they are not interacting with a human being. For all its advancements, primarily in the business sector, chatbots still have their flaws. There is definite room for growth and improvement.

Ms. Harini Jayasundara

Level 02
Department of
Industrial Management,
University of Kelaniya.

Memory Lane.

Walkthrough of Memories Made

Industry 5.0: The Future of Industrial Manufacturing

By **Himashi Silva** | Level 04

The majority of companies are still attempting to integrate Industry 4.0 concepts into their production processes, while the next industrial revolution, Industry 5.0, has already arrived. But before you start freaking out about the new kid in town, let's take a closer look at what Industry 5.0 is, why it exists, what technologies are used in it, and what it holds for the future of manufacturing.

What is Industry 5.0?

Let me walk you through the historical events that paved the way for Industry 5.0 to familiarize you with it. The development of the steam engine, which allowed for mass production, sparked the first industrial revolution. The second industrial revolution was brought on by the development of electricity, and Industry 3.0, sometimes known as the "digital revolution," was brought on by the widespread use of computers. With work digitization, which merged cyber technologies with actual manufacturing environments, the fourth industrial revolution entered the picture in Germany. Industry 5.0 is a new technology from the previous generation that is intended for intelligent, effective machinery. Putting it simply, the efficiency of industrial production is increased through the collaboration of humans and machines.

Industry 5.0, in the words of the European Union, "puts the welfare of the worker at the center of the production process and leverages new technology to bring prosperity that goes beyond jobs and growth while respecting the planet's production constraints." It basically means that Industry 5.0 is more concerned with well-being than welfare and with society's value rather than a commercial benefit. The three fundamental pillars of Industry 5.0 are human-centric, resilient, and sustainable. Let's delve further into these three pillars!

Human-Centric

Promoting talent diversity and employee empowerment at work will change the workplace from one where people serve companies to one where people serve organizations. This is what it means to be human-centric. This is more extreme than it might initially appear to be. Additionally, it fits in well with recent changes in the labor market. Finding, serving, and retaining talent has become a considerably bigger task than finding, serving, and retaining customers in many businesses and nations. If this trend persists, business strategy must provide it with a legitimate home, and that's what Industry 5.0 aims to do.

Sustainable

With the many concerns we currently have over climate change, the concept of sustainability hardly needs an introduction. This pillar emphasizes that in order for organizations to accomplish sustainable development goals, they must focus on all three Ps of the triple bottom line: the planet, people, and profit. However, fully integrating sustainability in a company's strategy involves much more than what is being done at the moment. Truly sustainable businesses put more emphasis on growing their positive impact than simply limiting their negative impact.

Resilient

Using flexible and adaptive technology, this pillar emphasizes resilience and agility. With the recent occurrences of the COVID-19 pandemic and the Russian-Ukrainian war and how these have influenced the global supply chain, there is no disputing the need for this in industrial manufacturing. Given that the majority of firms place more emphasis on agility and flexibility than on resilience, this represents as dramatic a change for an organization as the pillars described earlier. If we are to accept that resilience will actually be one of the three pillars of Industry 5.0, it follows that strategy's main emphasis will no longer be on growth, profit, and efficiency, but rather on building organizations that are "resilient," that is, able to anticipate, react to, and learn from any crisis quickly and methodically in order to ensure stable and sustainable performance!

Everything discussed so far has been related to Industry 5.0's conceptual aspects. Do you want to know more about its application of it? Let's enter the world of industry 5.0 technologies now!

Industry 5.0 Technologies

Cloud Computing

Efficient and effective innovation and scalable economics are offered by this technology. Through the usage of this technology, data is managed and stored on remote servers using the internet, and then it is viewed online. It offers apps, storage, and processing power together with on-demand computing services. The cloud infrastructure supports IoT edge platforms.

The platforms are used to control edge devices like various shop floor robots and autonomous robots. The industry accesses information from the local servers every day in order to manage crucial data. Industry 5.0 can minimize the amount of data transmitted to the centralized server. Wouldn't that be awesome?

Collaborative Robots

Restoring the human touch to development and production is a goal of Industry 5.0. It gives human operators access to advantages enjoyed by robots, such as their capacity for heavy lifting and technical precision.

Collaborative robots and Industry 5.0 both have important implications for the need for human contributions that can prolong current iterations. A new era in robotics and production is represented by collaborative robots and Industry 5.0. Collaborative robots and Industry 5.0 are the engine that can combine human ingenuity and craftsmanship with robot dependability and efficiency. This will ensure human-centric but machinery-involved production in industrial manufacturing.

Big Data Analytics

Enormous data analytics is a difficult process to evaluate big data to find data such as hidden patterns, market trends, and others. With various data sets, including structured and semi-structured data, it employs a cutting-edge analytical method. It requires using conventional methods to store and interpret enormous amounts of data. Businesses use big data analytics to take strategic decisions. Big data analytics may be quite helpful in tracking customer experiences and finding answers to issues in order to establish strong customer relations. Industry 5.0's sustainability element could benefit greatly from this.

Blockchain

It is a distributed and decentralized form of technology where recordings called "blocks" are used to store the transactional data in the digital ledger. It is a shared ledger that can make it easier to record transactions and keep track of assets inside a company network. Customers benefit from blockchain technology by being able to track orders, payments, production, etc. To prevent duplication of work and records in the database system, the network members maintain distributed ledger records of transactions. Despite the fact that blockchain has become a highly famous buzzword recently, it can tremendously help Industry 5.0 to shape the supply chains of the future industrial manufacturing.

6G network

It's a sixth-generation standard for creating wireless communication tools that can support cellular data networks. 6G networks are anticipated to give ultra-high reliability for Industry 5.0 by meeting the norms of the intelligent information society. To assure network connectivity, artificial intelligence techniques are applied to obtain mobility prediction solutions. With augmented and virtual realities, Industry 5.0 in the metaverse will be shaped by 6G technology, revolutionizing the manufacturing process. Isn't it great to see the prototypes in real life before actually making them? Well, 6G will help in making it into a reality!

Industry 5.0; The future of it

The idea of Industry 5.0 hasn't really taken off just yet. Businesses continue to invest substantially in Industry 4.0 or even earlier versions. But the developed countries in Europe have already started to move into the new era of Industry 5.0. Hopefully, Industry 5.0 will aid in reshaping manufacturing in the future to be more human-centric, durable, and sustainable, which will assist in revolutionizing current industrial manufacturing!

In conclusion, Industry 5.0 holds great promise for the manufacturing sector, so get your mind open to the innovation it will offer to your life.

Ms. Himashi Silva

Level 04
Department of
Industrial Management,
University of Kelaniya.

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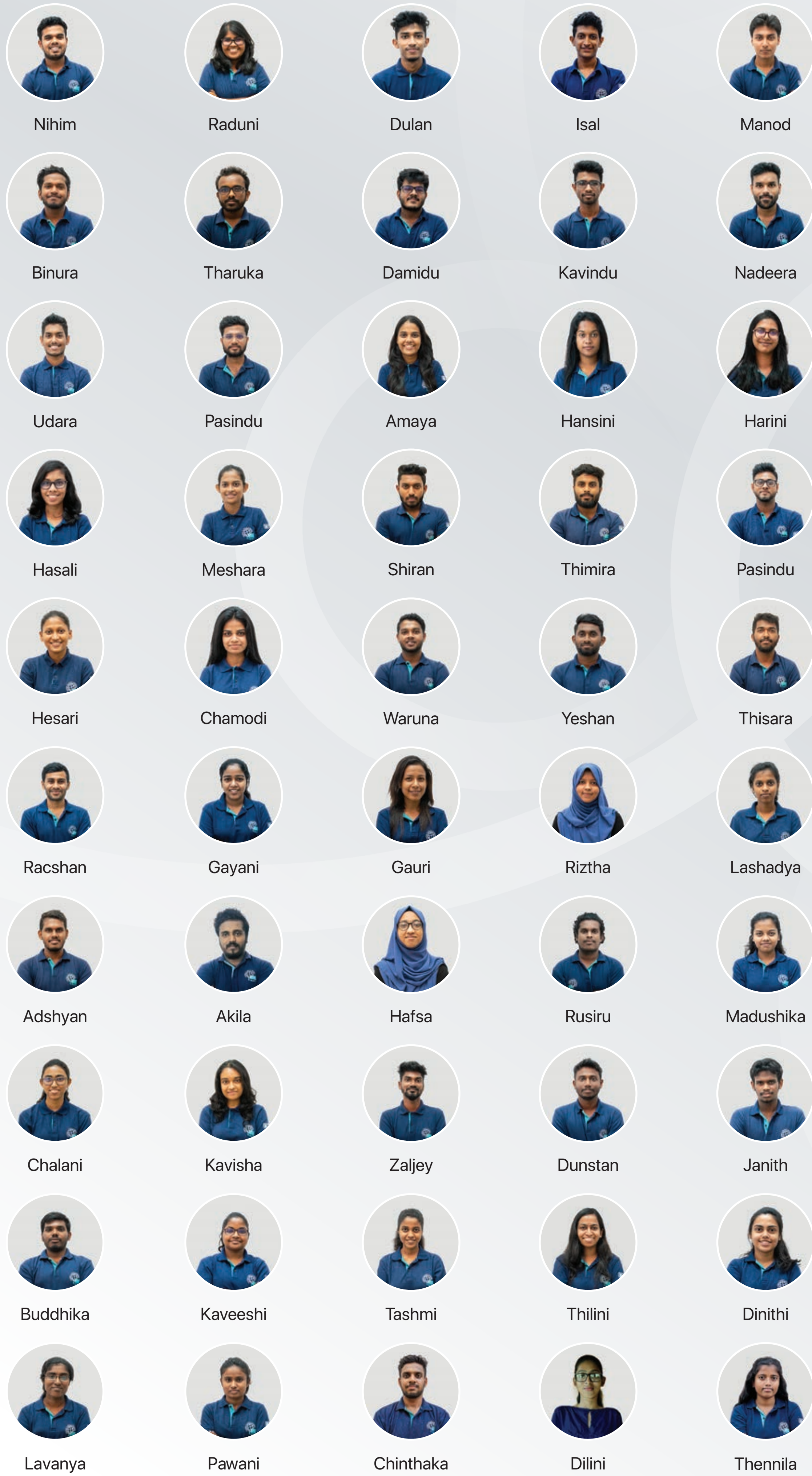

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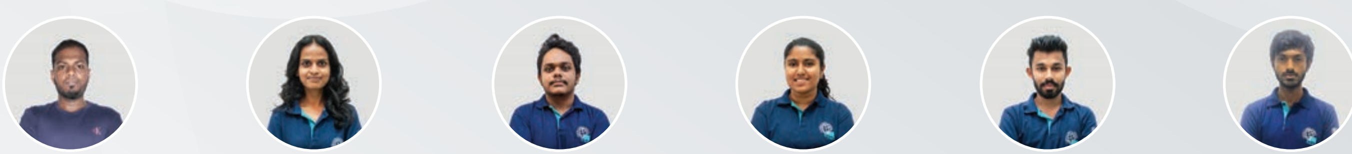

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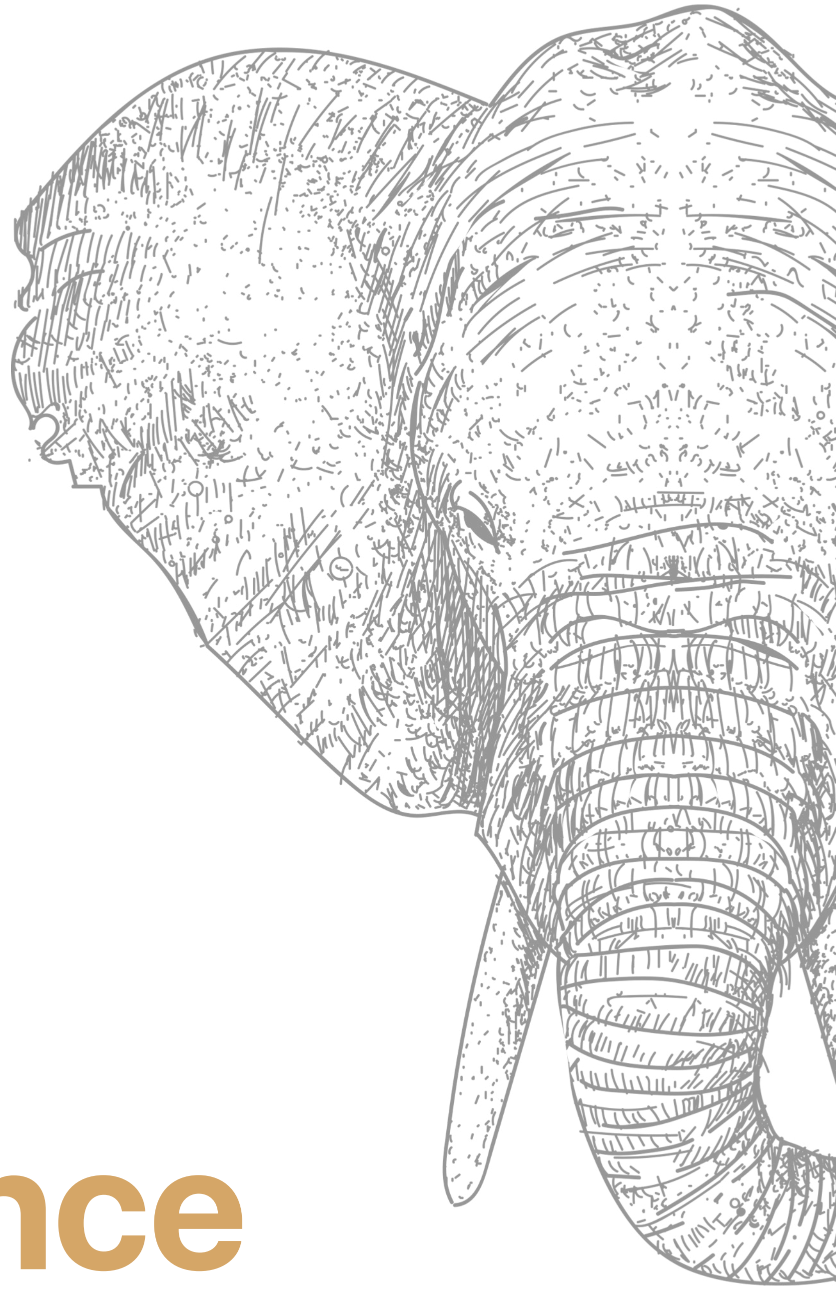

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